FlightGear Mirage 2000-5/2000D Manual

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1 FlightGear Mirage 2000 Documentation

Welcome to the documentation of the Dassault Mirage 2000⇗ for the FlightGear⇗ open source flight simulator.

The Mirage 2000 modelled here is mainly the -5 version (“Dash 5”/”tiret cinque”). The bi-place versions 2000D/N are also modelled to some degree (weapons, exterior 3d), but the cockpit remains for the moment the -5 version. There are also -5B and -5B backseater versions, but they deprecated, not actively tested and might therefore simply not work.

The Mirage 2000 is a French multi-role all-weather fighter aircraft from Dassault Aviation⇗. In the single-seater versions it is one of the best dogfighters⇗ of its generation and a far beyond visual range combat aircraft. The single-seater as well as the specialized double-seater versions are very capable in air-to-ground strikes.



1.1 This Manual

The goal of this manual is to be the most up to date source of information for pilots of this simulation model - in line with recent development of features. As such it documents how the FlightGear Mirage 2000 works - not how it works in reality!

This manual has been written in order to help you get to know the FlightGear Mirage 2000 and master flying and fighting with it. The French Delta Knife is not always easy to handle, but we hope the pages in this manual will allow you to become better at fighting with the modelled Mirage 2000 variants.

If you find rooms for improvement in this manual, then please contact us (see Section 1.6).

The old/original manual `Mirage 2000 manual.pdf` by Renaud "Wolf" Roquefort can be found in the "Docs" folder or accessed by this link. It is one of the sources of this manual and will eventually be replaced. However, it still contains information, which has not yet been migrated (e.g. engine and fuel systems, presentation of cockpit instruments, air-2-air interception procedures and combat, navigation). When there is overlapping information, then most probably this manual is better aligned with the current version of the model in FlightGear.

1.2 Realism

1.2.1 Sources of Information

This model in all variants is an interpretation of the real Mirage 2000 based on publicly available information (see Chapter 12) without any form of involvement from the original equipment manufacturers or any other entity with knowledge of unpublished/classified material of the original aircraft or its offensive/defensive systems.

Given the lack of authoritative information on military sensitive information, the realism of this model is by nature limited. Likewise, the goal of the simulation is for pure recreational activity - and therefore the simulation sometimes deliberately reduces and other times enhances real functionality/procedures to make the model more accessible to the casual simulation pilot. A lot of functionality / characteristics are "could-be" guesses good enough for gameplay purposes.

These remarks are especially relevant for the simulation of weapons/countermeasures/damage, which - apart from the natural lack of detailed documentation - are modelled as an opt-in to the FlightGear flight simulator and lack fidelity compared to reality. The fidelity and accuracy try to mimic what "could-be" real, but in the end just try to allow some recreational but still "realistic" game-play.

As such this manual and the FlightGear model are merely free (as in beer) spare-time personal projects for use in the FlightGear open source simulation community. This manual cannot be used to learn how to handle the systems of or to fly real Mirage 2000 variants.

1.2.2 Priorities

The priorities for modelling the Mirage 2000 in FlightGear are currently roughly as follows:

- Cockpit 3D modelling over exterior 3D modelling over weapons 3D modelling
- Digital displays over conventional instruments over switches
- Front seat over backseat
- M2000-5 over M2000N/D
- Compatibility with the latest stable FlightGear version (e.g. 2024.1.* as per year 2026) over backwards compatibility

1.2.3 Country variants

The modelled variants in FlightGear are mainly reflecting interior and exterior properties of the French and sometimes Greek aircraft - mainly due to easier access to information for the authors.

Currently there is no explicit modelling of export variants - neither in terms of 3D, specific weapons or systems, labelling of the cockpit etc. However, a larger set of liveries covering many countries and squadrons can be chosen.

1.2.4 Enhancements

If you have requests for changes / additions, which could further improve the simulation and thereby help to celebrate the Mirage 2000, then please contact us (see Section 1.6).

1.3 Other Documentation

Apart from this manual, there are other sources of documentation for the FlightGear version:

- The Mirage 2000 wiki page (not very well maintained) on the official FlightGear wiki.
- The main and original author of the FlightGear version has published some 5H1N0B1 YouTube videos (a bit aged).
- The excellently modelled FlightGear version of the F-16 has a very well written wiki, which also explains concepts (e.g. how a radar works).

RAZBAM has made a remarkable simulation of the Mirage 2000C variant for DCS. There is quite some documentation for it and an active forum, but it is an earlier variant for the Mirage 2000 (C) and therefore has somewhat other capabilities and a distinctively different cockpit.

1.4 Credits

- The authors of the model: Helijah (3D), 5H1N0B1 (Weapons, Systems), Richard Harrison (FDM), Leto, Josh Davidson (FCS/FBW/Autopilot), hardball, Domokos Juttner (Rudolf), F-Sig, J Maverick 16, legoboyvdlp, onox, ravenclaw_007, Niall “Salmon” Roe, Marsdolphin, R. “Wolf” Roquefort, Rick Gruber-Riemer
- The authors of the common weapons and radar systems for OpRedFlag, on which this model depends and which makes it possible, to use the plane in air and ground attack as part of military simulation in the OPRF (Open Red Flag) community.
- The authors of the FlightGear F-16 are a constant source of inspiration - and the level of sophistication of the model is what the M2000 is aiming for, but maybe never will get on par with.
- Helping hands for the creation of the manual: 5H1N0B1, CUDA22, I. “Interce” Klimov, SammySkycrafts, VooDoo3, Legoboyvdlp, Rudolf, Reaper

1.5 License

This FlightGear Mirage 2000 model including its documentation is licensed with GNU GPL⇗ version 2. See the licence information⇗ in the repo.

1.6 Contact and Contributions

Please do not hesitate to contact. The more information you provide (not at least links to published pictures, information etc.), the higher the chance that improvements can be made. Doing research and finding plausible details for how to improve the simulation is time consuming and contributions are very welcome by the maintainers of this model. You do not have to be a developer to help with progress!

- The currently most active developer can be reached by e-mail at rick AT vanosten DOT net.
- The GitHub repo⇗ provides the possibility to submit issues.
- The official FlightGear Forum⇗ has a dedicated thread for the Mirage 2000⇗.
- There is also a specific Discord server⇗ dedicated to the development of this simulation.

1.7 Conventions Used in this Manual

A "⇗" at the end of a link means that the link points to a site outside of this manual - otherwise to a different area within this manual.

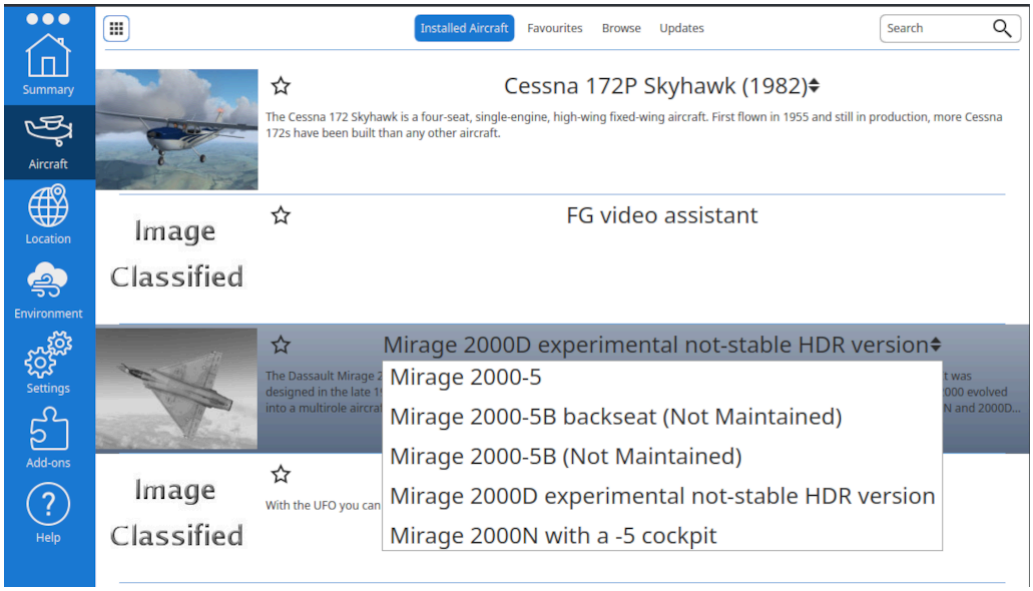
A reference to a specific asset in Chapter 12 is shown as `[ref: D01]`, where "D01" relates to the unique identifier in front of the reference's title.

2 Presentation of the Mirage 2000 Aircraft

Apart from restricted military information: the Mirage 2000 with all its variants is very well documented in books, articles and videos. Therefore, the Mirage 2000's history, technical characteristics etc. is not repeated here. Please use some of the resources listed in Chapter 12 instead.

3 Choosing a Variant to Fly

There is a set of variants to choose from in the FlightGear Launcher as shown in the picture:



Stable versions working in FlightGear version 2024.1:

- `2000-5`: the single seater fighter variant with 5 displays
- `2000N`: the double-seater strike variant with ground-attack weapons (it does re-use the -5 cockpit and the back-seat is not usable and some -D variant weapons are also available)

Not maintained versions which might work in FlightGear version 2024.1:

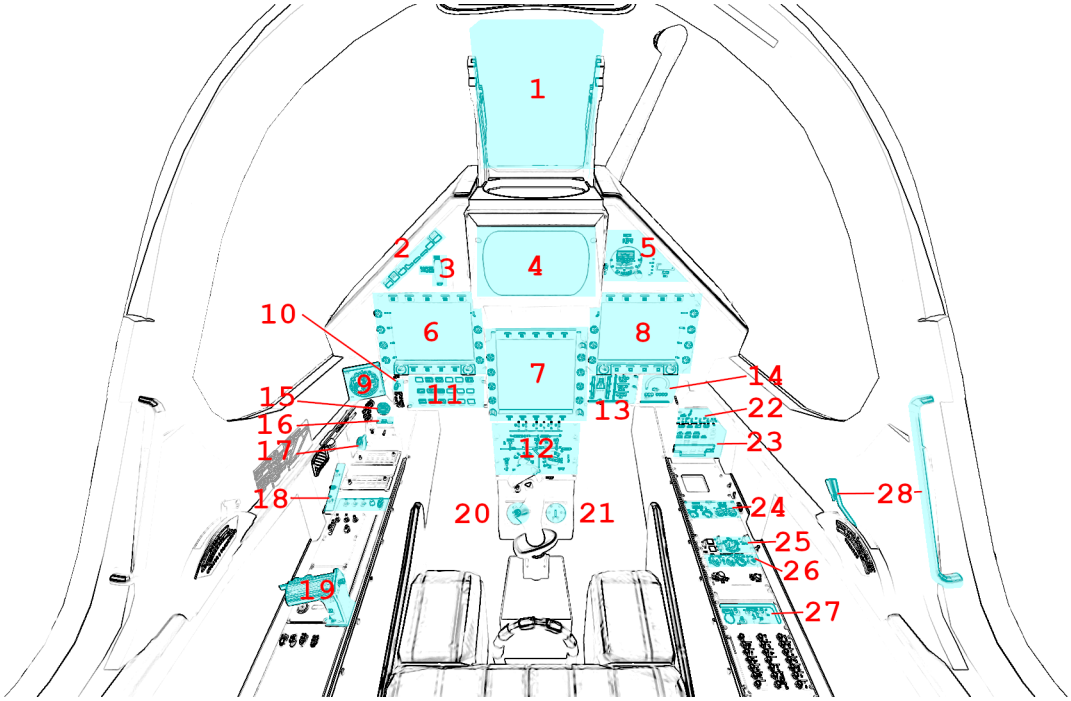
- `2000-5B`: the trainer version for the -5 variant
- `2000-5B backseat`: sit in the backseat of the -5B over FlightGear multiplayer

Experimental versions only working with the next/development version of FlightGear - not stable/no promises:

- `2000D`: the double-seater strike variant with ground-attack weapons (it is getting its own cockpit - at least in the front)

4 Cockpit instruments

4.1 Overview



1. Head-up display (HUD) - French: visualisation tête haute (VTH)
2. Autopilot panel
3. Elevator trim
4. Head-level display - French: visualisation tête moyenne (VTM)
5. Physical flight instruments
6. Left Multi-function display (MFD) - French: écran multifonction
7. Head-down display - French: visualisation tête basse (VTB)
8. Right MFD
9. Clock
10. Master Arm switch
11. Weapon command panel
12. Transponder
13. Fuel indicator panel
14. Engine indicator panel
15. External tank jettison button
16. Gear indicator
17. Gear lever
18. Light switches
19. Throttle stick
20. Elevon angle indicator
21. Cabin altimeter (non-functional)
22. Power switches

23. Alert panel
24. ILS & TACAN panel
25. Air conditioning panel
26. Interior lighting
27. Engine control panel
28. Canopy handle & lock

Please note that this cockpit model is not complete. Only instruments that have been implemented are highlighted.

4.2 HUD [1]

Please refer to Section 5.1.

4.3 Autopilot Panel [2]

Please refer to Chapter 7.

4.4 Elevator Trim [3]

This instrument allows for the angle of the elevons to be manually offset in order to adjust the rotational angle of the aircraft along the pitch axis. It can be controlled by scrolling while hovering one's mouse over the wheel.

Please note that the influence of this instrument is rather low, and high-G manoeuvres should rely on the stick input more so than the trim.

4.5 VTM [4]

Please refer to Section 5.2.

4.6 Physical Flight Instruments [5]



These instruments serve as backup for the digital one, as well as when the left MFD is not on the `EADI` tab.

4.6.1 Angle of Attack (AoA) Indicator [5.1]

This gauge enables the pilot to see the angle of attack of the aircraft. One dot equals to 5° up until +15°, then 6.7° approx up until 35°. Negative AoA will not be shown.

While the onboard computer always attempts to keep the Mirage out of a stall (>25° AoA) and pulling on the stick regardless of AoA is not an issue, avoiding a tailstrike (>14° AoA) on takeoff and landing is of utmost importance. As such, should the HUD not function, this indicator will have to be taken into account on these instances.

Note that this indicator might not be visible depending on your view position. Moving your head to the side will allow you to see it.

4.6.2 Compass [5.2]

This indicator will display the current heading in degrees. Depending on the setting of the VTB, it will use either the True North or the Magnetic North.

4.6.3 Airspeed Indicator [5.3]

This indicator will display the current airspeed in knots as well as the current mach below.

4.6.4 Attitude Indicator [5.4]

4.6.5 Variometer [5.5]

This indicator will show the vertical speed of the aircraft at low values. Each horizontal bar corresponds to 500ft, with a shown range between -2,000ft/min and +2,000ft/min.

4.6.6 Altimeter [5.6]

This indicator will show the altitude above sea level, adjusted to the pressure set in the `EADI` tab of the left MFD.

4.7 Left MFD [6]

Please refer to Section 5.4.

4.8 VTB [7]

Please refer to Section 5.3.

4.9 Right MFD [8]

Please refer to Section 5.4.

4.10 Clock [9]

This clock will display the time in UTC.

4.11 Master Arm Switch [10]

This switch sets all weapon stations as well as the onboard cannon on live fire mode, and must be toggled on to fire any type of armament. It is off by default.

This switch should only be turned on in combat situations.

4.12 Weapon Command Panel [11]

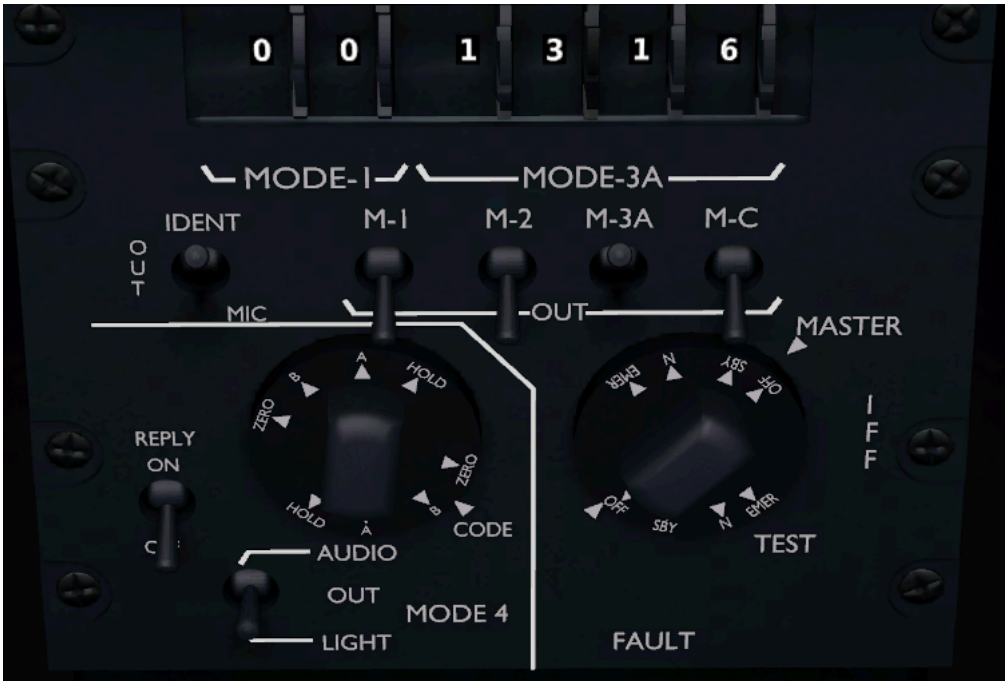
[Not implemented]

4.13 IFF Transponder [12]

The Mirage 2000 IFF transponder panel includes both civil and military functionality. A Transponder assists air traffic control to identify an aircraft. See Chapter 9 for information on IFF.

According to the RAZBAM M2000 DCS manual [ref: D04] the modelled transponder is a `NRAI-7C`.

NB: the generic FlightGear civil transponder can be accessed using menu item `Radio Settings` (`Key: F12`) in menu `Equipment`. There is a description in an FG wiki article and it will therefore not be described further here. However, you should use the Mirage transponder panel instead of the `Radio Settings`, because the two are not fully compatible and the 3D panel does not consistently update by settings through the dialogue. If you nevertheless use the built-in dialogue, then make sure to use the "Close" button on the dialogue to close it - otherwise the `IDENT` button will remain pressed if you have pressed it.



4.13.1 Upper Part of Panel

The top left 2 wheels `Mode-1` do not work. The top 4 right wheels `Mode-3A` are used to input the civil transponder code (which corresponds to the value in the `Radio Settings`).

The `IDENT` switch is spring loaded and can be used to respond to civil interrogations.

The military interrogation switches `M-1` and `M-2` do not work.

The `M-3A` switch works for civil identification interrogation (mode A) and the `M-C` (mode C) works for civil altitude reporting interrogation. Only one of them can be active, but both can be off.

The modelled transponder NRAI-7C does not support interrogation mode S.

4.13.2 Mode-4 Subpanel

The position of the bottom left switch `Reply` determines whether the IFF interrogation functionality is powered `ON` or `OFF`.

For an interrogation to be answered, several things need to be in place:

- The `Reply` switch is `ON`.
- The selector is on either `A` or `B`. The IFF channel value for A and B are set in menu item `Mission Preplanning` accessed through menu `Mirage 2000` (if damage is on, then it can only be done on ground).
- The master selector is on `N`.

Setting the selector to `ZERO` will zero out the IFF channel A and B values. Know what you are doing, because to new

values need to be input through the Mission Preplanning dialogue, which might not be available in flight!

According to [ref: D04] position HOLD "prevents the zeroizing of the MODE 4 IFF codes when the IFF transponder is turned off." This is not implemented: there is not zeroizing modelled when the aircraft looses power etc. If the selector is on HOLD then no response will be given.

The Audio-Out-Light switch is not implemented.

4.13.3 Master Subpanel

Changing the selector to N automatically sets M-3A if neither M-3A nor M-C is set. Changing the selector to EMER automatically sets M-3A no matter the setting for M-3A and M-C.

N (neutre) means that civil interrogations for M-3A or M-C are enabled, military interrogations for mode 4 are enabled and military interrogations for modes 1 & 2 are enabled (but modes 1 & 2 are not implemented).

NB: even if the selector is on N: an actual interrogation still needs to be replied to using the IDENT switch for civil interrogations or the Reply switch for military interrogations in mode 4.

4.13.4 IFF Interrogator Panel

In the real Mirage aside the transponder panel there is also an IFF interrogator panel on the right console near the ILS & TACAN panel. This is not implemented - i.e. you cannot see it in 3D and you currently cannot interrogate other aircraft from the Mirage.

4.14 Fuel Indicator Panel [13]



Displays the remaining fuel in kg (Note that the amounts in the Equipment -> Fuel and payload menu are displayed in lbs, with 1 lbs = 0.45 kg or 1 kg = 2.2 lbs approximately).

GAUGE will be the total amount of fuel in the internal tanks (feeding system aside).

REMAIN will also account for the feeding line and the external tanks.

BINGO is a value that can be manually set in the Mirage 2000 -> Configuration menu. It should be set as the minimum fuel required to return to base (RTB) - having less than this will cause the fuel indicator to flash red, signalling the pilot to urgently RTB. By default, it is set at 480kg, which is a rather low value.

The white Mirage-shaped indicator on the left displays in white sections of the fuel system that are not empty, and in black if they are devoid of any fuel. The upper two rectangles account for both the forward and backward fuselage tanks of each side, and the pentagons for the wing internal tanks. When taking external tanks, they will be displayed as white disks below the aforementioned shapes.

Note that the fuel system will first attempt to empty the external tanks before using the fuel contained in the internal system. If the tanks are jettisoned, the fuel flow will automatically switch to internal tank feed.

4.15 Engine Indicator Panel [14]

This indicator will display information about the engine's speed and fuel consumption.

The top value ("N%") displays the ratio of the engine's speed to its maximum military power (i.e. without afterburners). In the idle state, it should be stable at around 47%, and at maximum military power at around 96%. Using afterburners will push this value above 100%.

The bottom left value shows the estimated fuel consumption per minute in kg. Note that this is an instantaneous estimation, which means changes in altitude, speed, etc, will affect it.

The bottom right value displays the number of engine rotations per minute (RPM).

4.16 External Tank Jettison Button [15]

Self-explanatory name. This does not jettison weapons attached to the pylons of the aircraft.

Jettisoning the tanks should only be done in dogfight situations or in case of emergency - they come from taxpayers' money, after all.

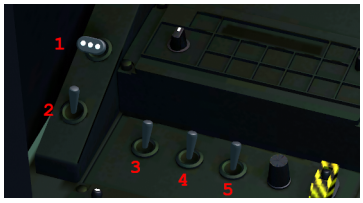
4.17 Gear Indicator [16]

Will display three green downward arrows when the gear is fully lowered. These indicators will disappear once the gear is moving or retracted.

4.18 Gear lever [17]

Lowering the lever will lower the gear, and raising it will retract the gear. Using the `g` and `G` keys (retract and extend gear respectively) will do the same, but also switch to the `NAV` and `APP` modes respectively.

4.19 Light Switches [18]



1. Taxi/landing light. Off by default.
2. Dorsal flash lights. On by default. Should be manually turned off after startup.
3. Formation lights (stripes on the sides of the fuselage and tail). On by default.
4. Tail position lights. On by default.
5. Wing position lights. On by default.

4.20 Throttle Lever [19]

Cannot be moved via the mouse, only with `Key: PageUp` and `Key: PageDown`. Afterburners are enabled at 90% of the lever's maximum extension.

NB: this value is different in a real Mirage 2000, where it lies at 75%.

4.21 Elevon Hydraulic Pressure Indicators [20]

Not functional.

4.22 Cabin Altitude Indicator [21]

Not functional.

4.23 Power Switches [22]

The red power switch toggles the battery on/off. The battery should be on at all times when the engine is running. Off by default.

The grey switches toggle all the alternators at once. They should be on at all times when the engine is running. Off by default.

4.24 Alert Panel [23]



[Default state of the alert panel when launching the simulation]

The warning lights should all be off in a normal situation (save for the parking brake when stopped on the ground). Depending on the severity of the warning, you might have to review the cockpit's switches, carry out an emergency landing, or eject. Their following codes are as follows:

Abbrev	Alert
BATT	Battery off
TR	Alternators off
ALT.1	Alternator 1 off
ALT.2	Alternator 2 off
OIL	Oil pressure too low
T7	N/A
CMPTR	Computer failure
RPM	RPM too high
VSD	N/A
LP	Fuel flow irregular
LLP	Left fuel pump off
RLP	Right fuel pump off
HYD.1	1st hydraulic system failure
HYD.2	2nd hydraulic system failure
EMG HYD	Emergency hydraulic system failure
EP	
BINGO	Fuel lower than `BINGO` value set
CAB P	Cabin pressure too low
TEMP	Temperature too low
OX REG	Engine oxygen flow irregularity
5mn OX	Low oxygen (5min remaining) (not implemented)
HA OX	Cockpit oxygen system failure
PITOT	Pitot tube failure
DC	N/A
CONDIT	Air conditioning failure

CONF	N/A
GAIN	N/A
SCOOP	NACA scoop failure
FLT ENV	Flight envelope failure (aircraft no longer flyable)
S CONES	Supersonic cone failure
EL B UP	N/A
AOA	Too high AoA
SLATS	Slats failure
MAN	N/A
T/O	N/A
PARK	Parking brake enabled
AP	Autopilot failure

4.25 ILS / TACAN Panel [24]

The VOR.ILS value can be tuned to an airport's instrumental landing system frequency in order to help with the landing. When in APP flying mode and if the ILS is enabled, you will be able to visualise the corresponding airport's runway in the HUD. The left knob changes the frequency by 1 MHz and the right knob by 0.05 MHz. The left knob's real function to set the system on (M) or off (A) is not implemented. Neither are the rights knob's testing functions.

Alternatively the frequency can be changed using menu item Radio Settings (Key: F12) in menu Equipment.

The TACAN allows the pilot to change the numerical value of the TACAN channel. The left knob (in the centre) changes it by 10 and the right knob by 1. To switch between the X and Y band the left knob's border can be used. The TACAN operational mode cannot be changed in any way.

Alternatively the TACAN channel can be changed using menu item Radio Settings (Key: F12) in menu Equipment.

4.26 Air Conditioning Panel [25]

The COND switch toggles the air conditioning inside the cabin. Off by default.

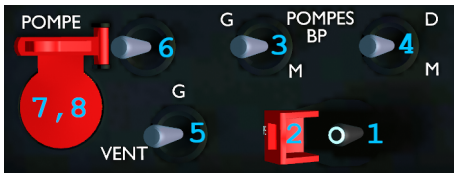
The knob to its right allows the pilot to set the desired air temperature of the air conditioning. Pointing the hand of the knob towards upper half will make use of the automatic temperature regulation system, while the lower half will switch to manual control of the temperature of the airflow (and is not advised). Each movement of the hand (in-sim) will offset the temperature by 1.33°C from the default temperature (22°C AUTO). Turning the knob to the right makes the temperature cooler, and to the left makes it warmer. It is advised to set the temperature to around 17-18°C AUTO.

The DESEMB switch toggles the windshield fog removal (French: désembuage). Off by default. It is highly advised to turn it on for medium-to-high-altitude flights.

4.27 Interior Lighting Panel [26]

Controls the cockpit lights.

4.28 Engine Control Panel [27]



[Default state of the engine control panel when launching the simulation]

Panel used for starting up the engine.

In order of the startup sequence:

1. Engine cut-off switch. Enabled by default.
2. Cover of the cutoff switch. Closing it disables the cutoff switch. Open by default.
3. Left fuel pump switch. Off by default.
4. Right fuel pump switch. Off by default.
5. Startup mode switch. Off by default.
6. Pump BP switch. Off by default.
7. Starter button cover. Closed by default.
8. Starter button. Pressing it for a few seconds gives the engine the necessary rotational speed to keep turning on its own.

4.29 Canopy Handle & Lock [28]

Clicking the canopy handle will switch between almost closed and fully opened states. When the canopy is almost closed, clicking the locking lever will fully close and secure the canopy. The canopy is fully opened by default.

Pressing `d` twice equates to clicking the canopy handle and the locking lever (and thus closes the canopy from the default state, or opens it completely if it is closed).

5 The Mirage 2000-5 Display System

The Mirage 2000-5 has its name from the fact that it has 5 main displays:

- Pink - Head-up display (HUD) - French: visualisation tête haute (VTH)
- Amber - Head-level display - French: visualisation tête moyenne (VTM)
- Red - Head-down display - French: visualisation tête basse (VTB)
- Blue - Left and right Multi-function display (MFD) - French: écran multifonction



Currently all simulated variants use this system, because they use the same cockpit. In real life the cockpit of e.g. the -D variant looks quite different and does neither have the right MFD nor does it have the VTM.

5.1 HUD

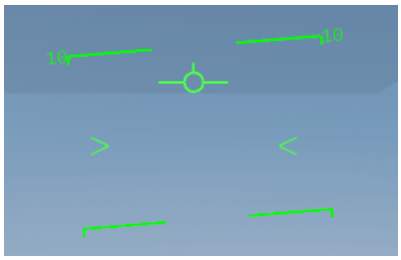
NB: Some description of information displayed in the HUD is included in other sections.

5.1.1 Acceleration Vectors

The acceleration vectors (aka. chevrons) indicate the current acceleration. If they are displayed above the flight path vector (the little airplane), then the aicraft is accelerating (gaining speed). A specialty of the M2000-5 version is that the chevrons are double when the afterburner is on, otherwise single.



If the chevrons are displayed below the flight path vector, then the aircraft is currently slowing down.



5.1.2 Ground Sub-mode

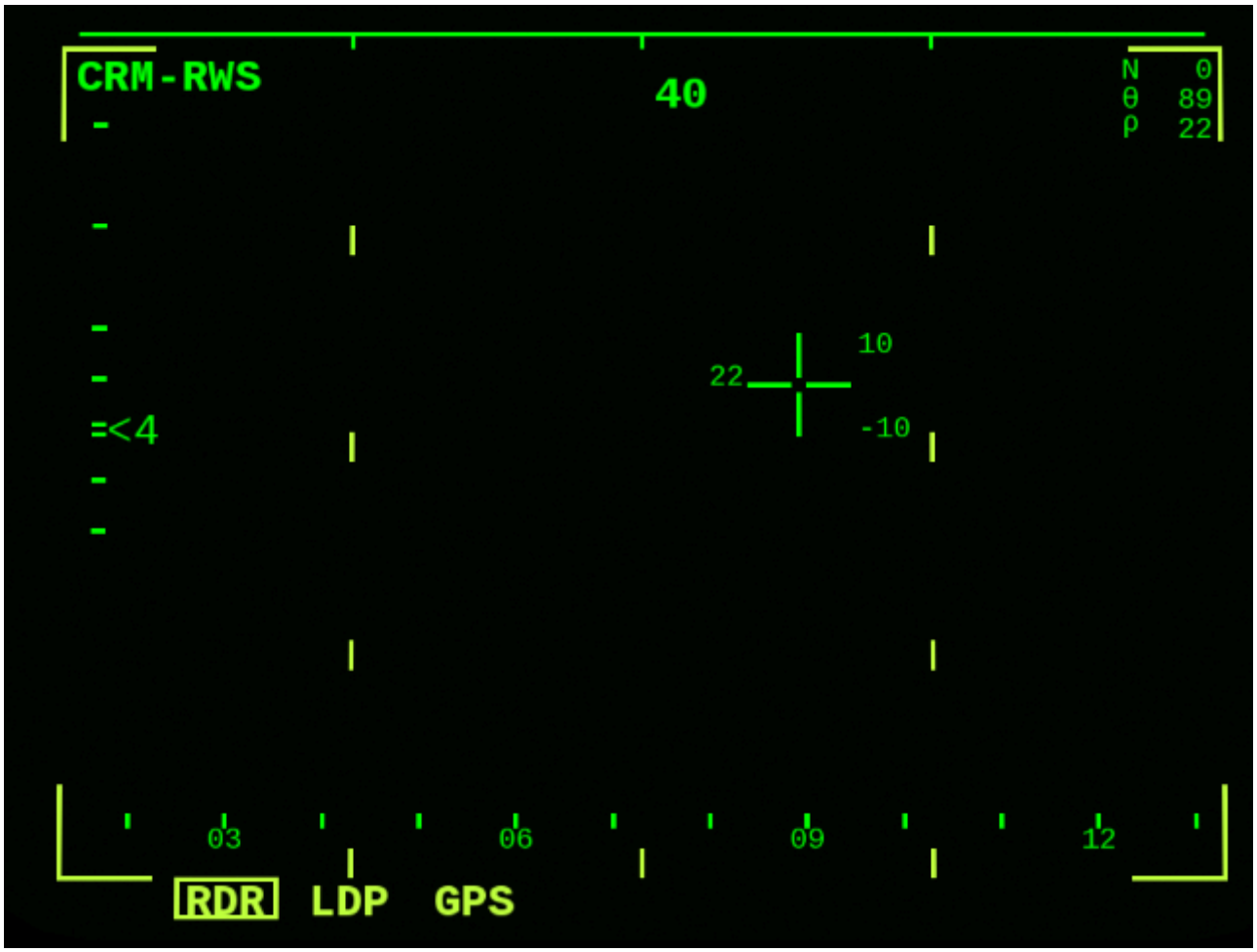
This mode clears much of the screen and displays 2 unique indicators:

- A box showing the longitudal acceleration in Gs (can be negative when slowing down). It is mostly used for checking the engine performance at take off. The values are typically well below 1.0 and depend on the thrust of the engine, the air resistance and the weight.
- An inverted `T` - the rotation index. Putting this line on the horizon at rotation speed will set the aircraft at a pitch angle of 13 degs, which is the desired angle to take off and avoids tail strikes. During landing and aero braking when only the 2 main wheels are on the ground the inverted `T` can help to keep the angle of attack at around 13 degs.



5.2 VTM

This is a specialty of the Mirage 2000-5. Being positioned just below the HUD, this screen allows the pilot to see radar related information without moving the head. Like the HUD the VTM's focal point is perceived to be at infinity.



Most radar pictures are displayed in a B-scope (see picture above), the sea and ground radar modes are displayed as Plan Position Indicator - cf. Radar Displays on Wikipedia.

The top left corner shows the radar main mode (`Key: Q` to cycle) and the radar sub-mode (`Key: ctrl-q` to cycle). In the top middle the radar range in nm is displayed (`Key: R` to increase, `Key: E` to decrease). The box in the top right corner shows data from the cursor: `θ` shows the the heading of the cursor, `P` shows the distance of the cursor on the radar.

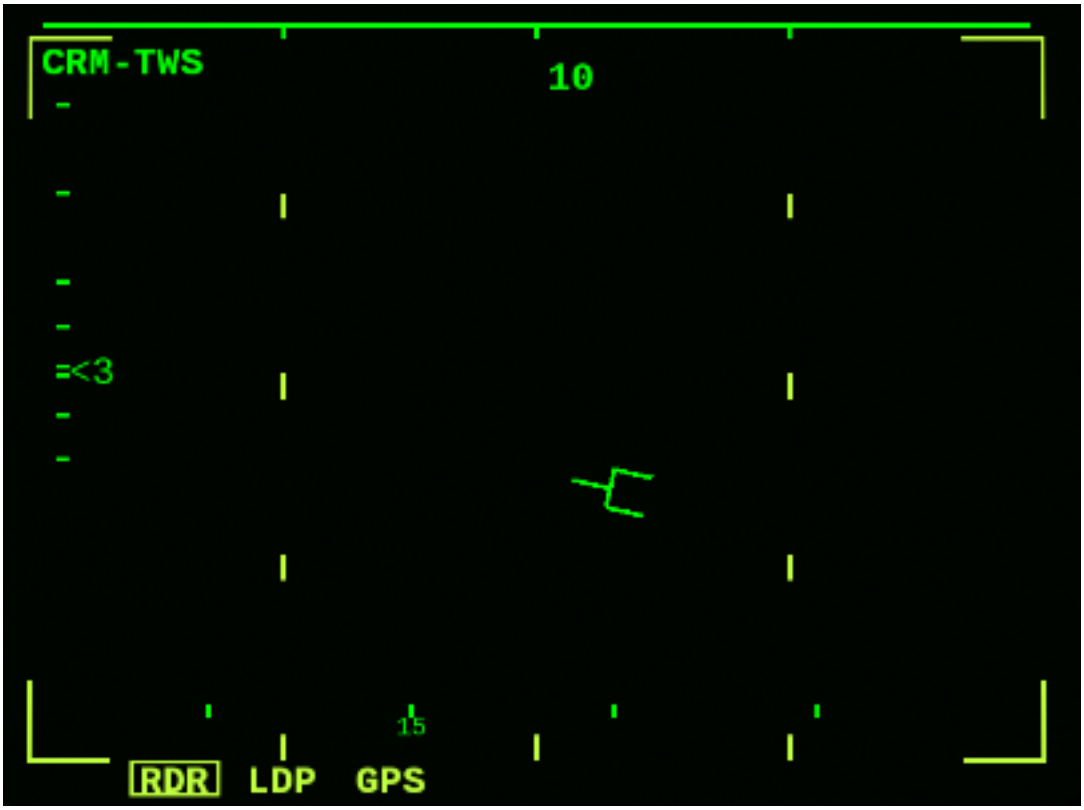
The cursor can be moved with arrow keys on the keyboard or using a binding to a joystick. On the left side of the cursor the radar distance is shown, on the left side the top and bottom altitudes being scanned by the radar (in ft).

At the bottom of the screen there is a heading scale (showing true of magnetic North depending on the setting done on the VTB). At the bottom on the left the three main weapon guidance modes are displayed: radar (RDR), laser designation point (LDP), GPS. The currently used guidance mode is displayed inside a box and can be changed using `Key: M`.

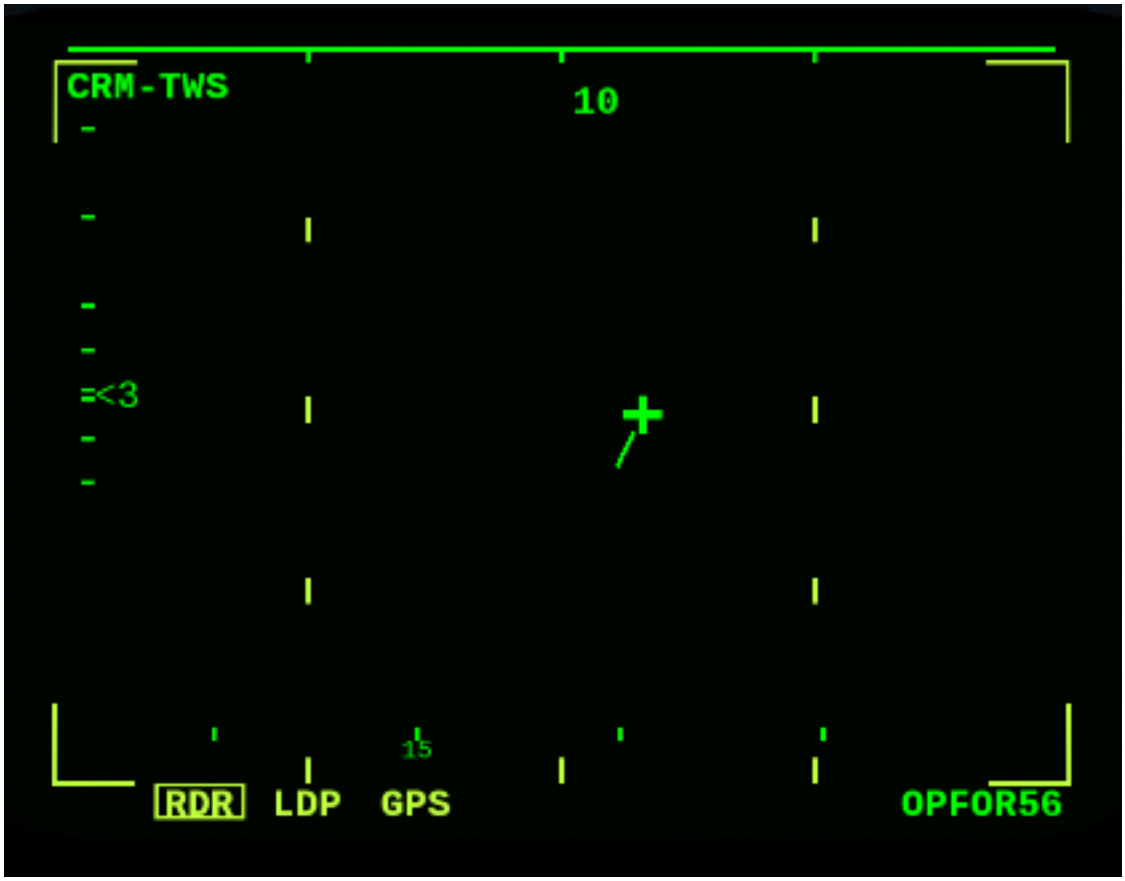
The scale on the left side shows the radar antenna elevation / radar pitch. The number shows the number of bars (vertical scanning of the radar). The antenna elevation can be changed (`Key: i` for up, `Key: I` (capital i) for down, `Key: Y` for level).

5.2.1 Air-to-Air

A flying target is shown as an open rectangle with a line showing the direction the target is flying at (the longer the line, the faster).



In TWS mode a target can be selected using `Key: y` (and deselected with `Key: ctrl-y`). When the target is selected, then the display is changed to a small cross (still with a line indicating the direction) and the targets identifier is displayed in the lower right corner.



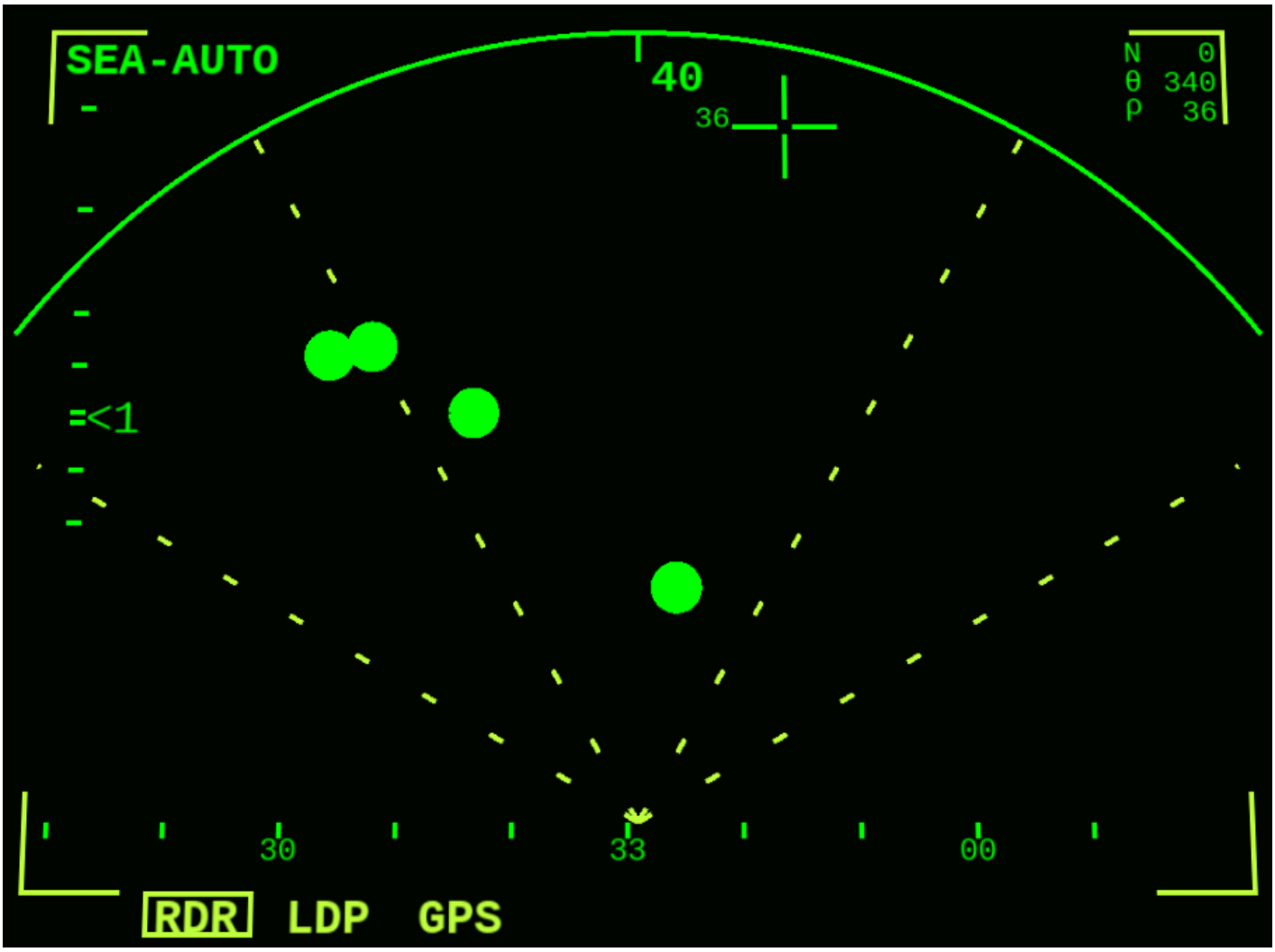
To select a target in another air-2-air or air-2-ground or air-2-see mode, the cursor must be moved over the middle of the target and then designated using `Key: l` (lowercase L).



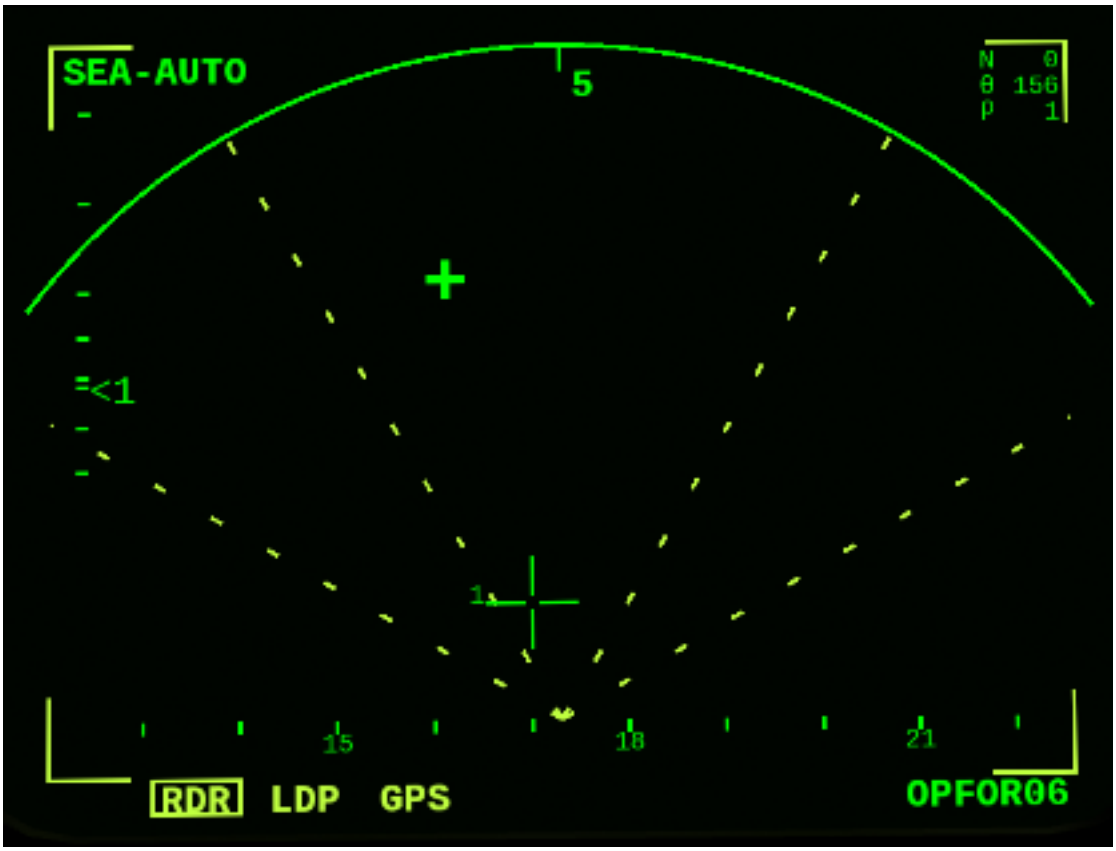
5.2.2 Air-to-Sea

Like all ground modes the radar picture is displayed as PPI.

In air-2-sea mode the target is selected and designated using the cursor. A not designated target is shown as a filled circle.



When the target is selected/locken, then it is shown as a cross. There is no information about direction / speed displayed.

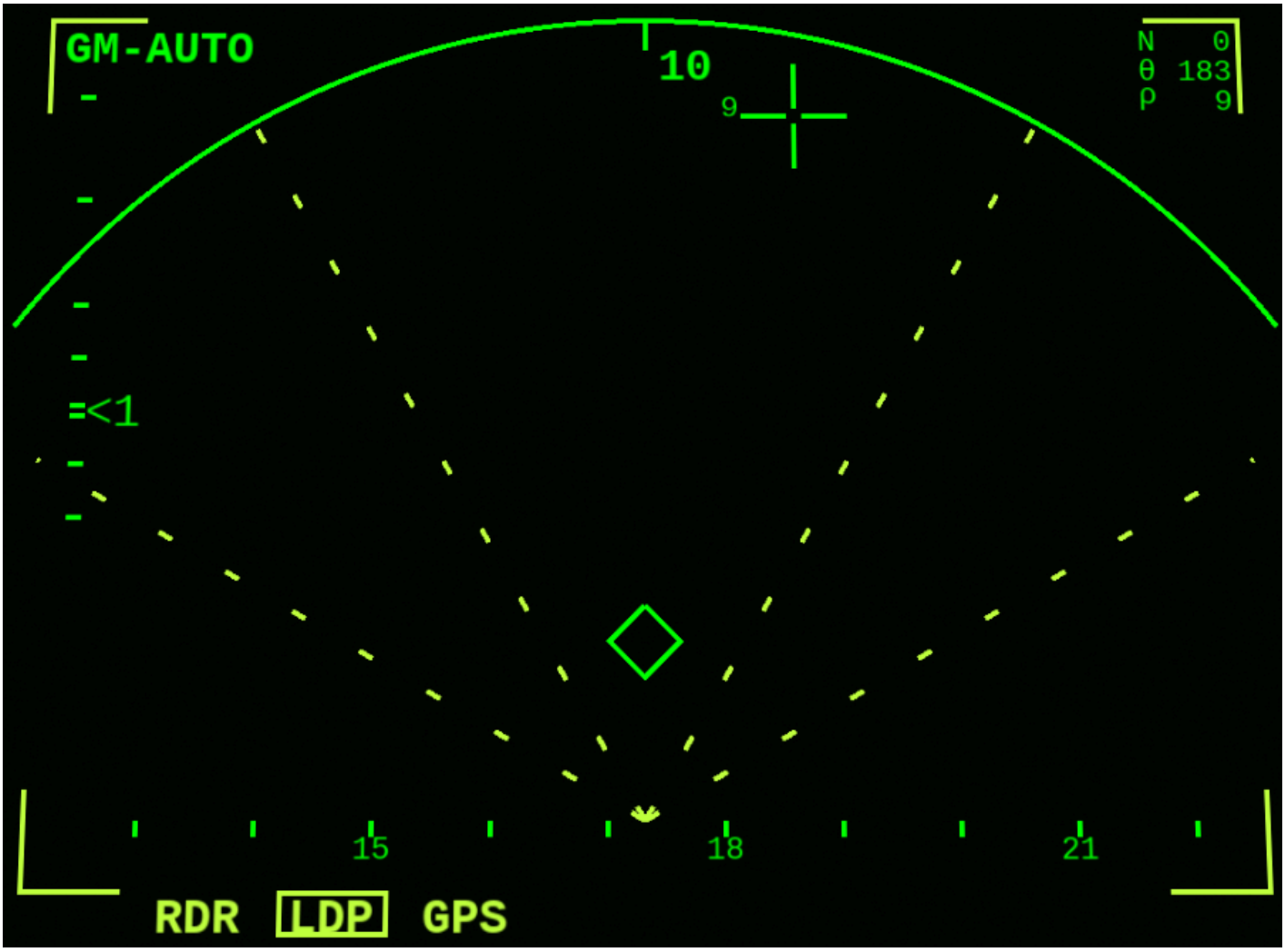


5.2.3 Air-to-Ground

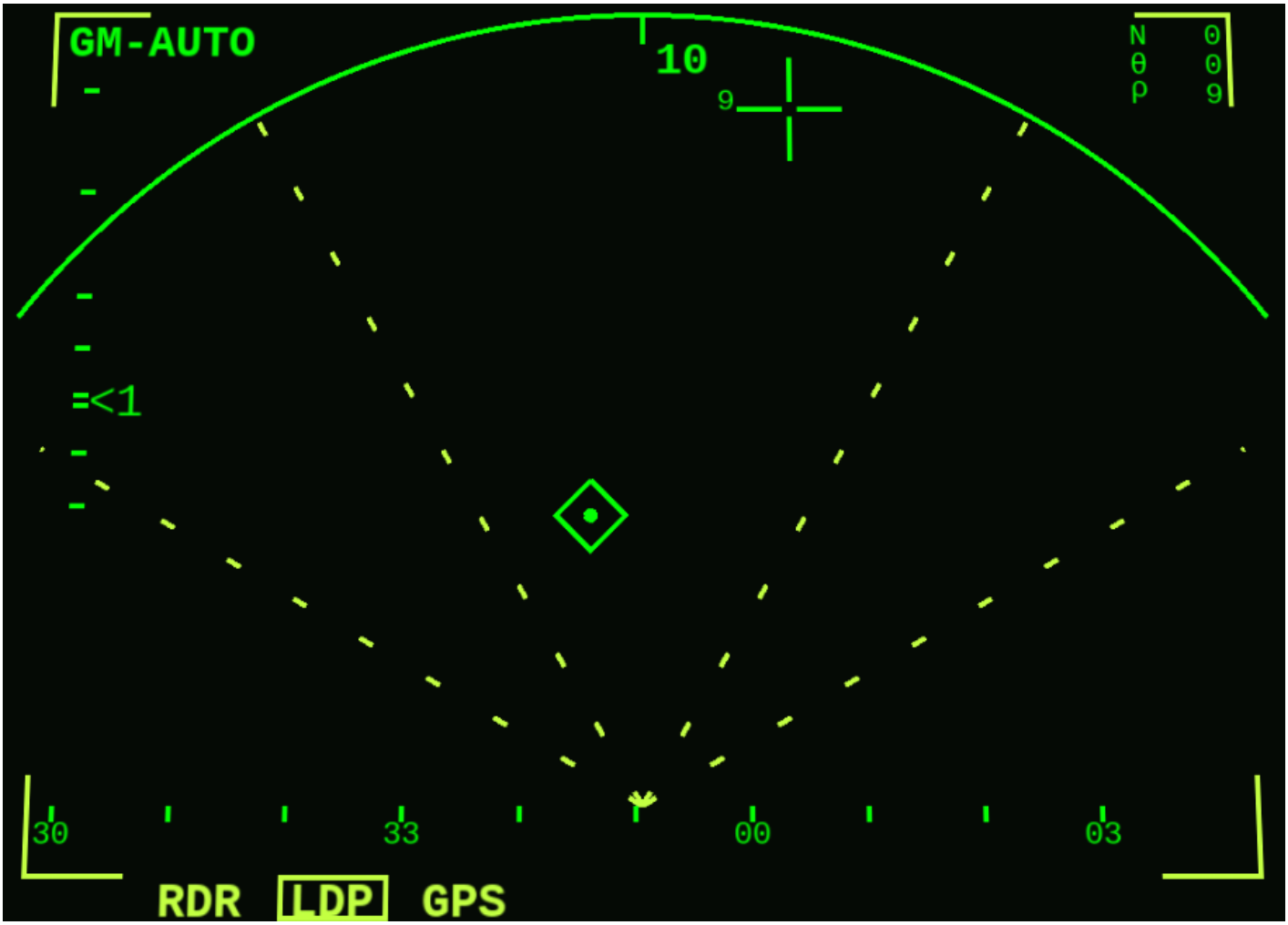
The following picture shows a few targets in ground mode (like for sea targets they are displayed as filled circles). Notice that the antenna elevation has been lowered a few degrees (radar pitch scale at left side). It is easier to get ground contacts when flying e.g. level at 1000 ft above ground and a shallow antenna elevation then flying high with or without pitch.



When a target has been marked a spotted, then it appears as a diamond (not filled) in the display. LDP guidance mode needs to be on.



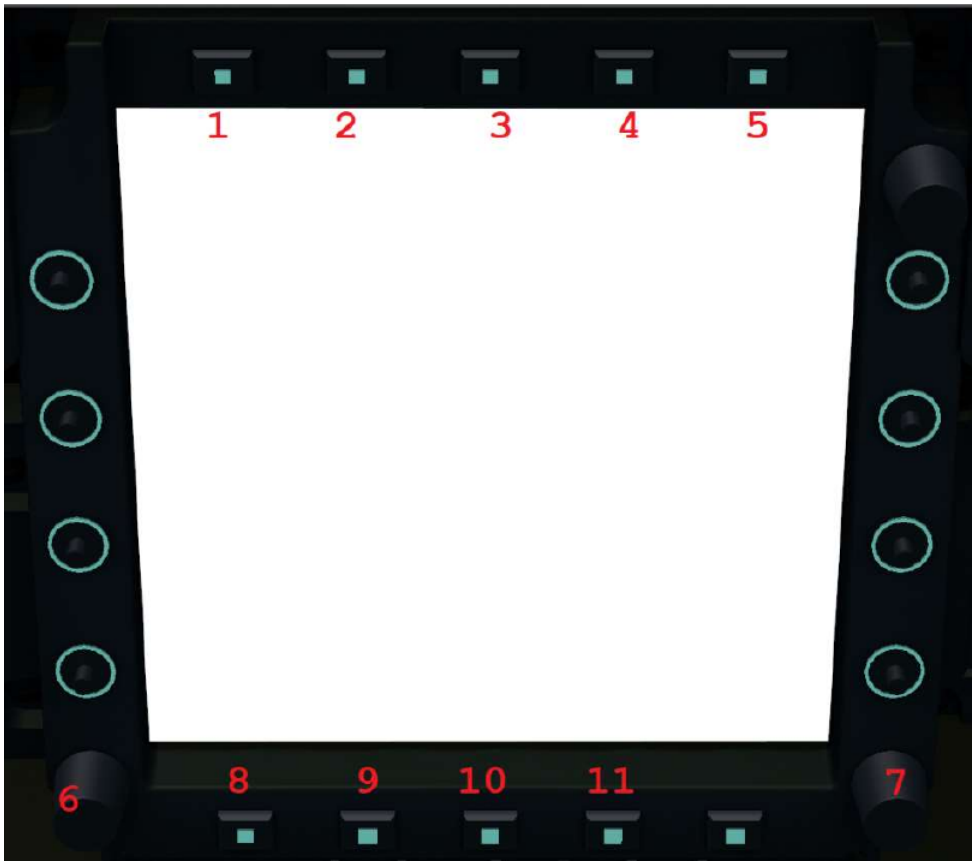
When the spotted target is then designated (“locked”), the diamond gets a dot in the middle:



5.3 VTB

The head-down display is currently basically a copy of a Navigation Display you would find in an Airbus - and thus not yet implemented as per the original.

The buttons around the screen influence settings as follows:



1. Show/hide airports
2. Show/hide waypoint data
3. Show/hide waypoints
4. Show/hide position points
5. Show/hide weather data
6. On/off button
7. Set range in nautical miles (distances: 10; 20; 40; 80; 160; 320)
8. Cycle pages: PLAN, VOR, APP, MAP
9. Toggle liquid crystal display/cathode ray tube display (does not really do anything)
10. Toggle centre ND
11. Toggle true/magnetic heading

5.4 Left and Right MFDs

From the hub (see illustration) you can access all available pages.



You can use `Key: Home` to get to the next page on the left MFD and `Key: END` to get to the next page on the right MFD. The page sequence is the same as the sequence in this chapter.

To change back to the hub page, use the second button from left (amber frame) on the button row at the top of the MFD. The text below the button shows the current page name.

To go directly to the RWR page, use the first button from left (green frame) on the button row at the top of the MFD.

Use the mouse wheel to push the toggles (yellow frame) on the left and right side of the MFD up and down. Only those toggles work, which have a text associated on the screen. Alternatively, use left-click to push the toggle up and right-click to push the toggle down.

You can focus the view with `Key: n` to better see the left MFD respectively `Key: N` to better see the right MFD. Use `Key: ctrl-n` to go back to the default view.

NB: the available pages on the left and right MFD are the same.

5.4.1 Electronic Horizontal Situation Indicator (EHSI)



The FlightGear wiki has a quite comprehensive page about Radio Navigation⇗ with a fair amount of links.

Navigation Modes

The navigation modes can be selected on the left side of the screen (`FMS` is active by default if none of the others is highlighted):

- VOR (incl. ILS) for NAV1 and NAV2
- DATA
- TAC (TACAN)
- FMS (Flight Management System - waypoints)

You can change between NAV1 and NAV2 with the button at the right of the top botton row - unless NAV* is displayed because you are in another navigation mode than VOR. In that case you can change to NAV1.

Compass

The compass rose is turning as the airplane turns. The current heading is at the top. You can toggle between true heading (TH) and magnetic heading (MH) with the knob on the left side.

Current Selected Course

The big green arrow shows the current selected course for either NAV1 or NAV2. NAV1/NAV2 can be toggled with a button on the right side of the top button row. The course can be corrected with the +/- knob on the right side: + changes by +10 degrees, minus changes by -1 degree.

Heading Towards Selected Destination

The blue arrow shows the direction towards the selected destination, which depends on the mode (VOR, DATA, TAC, FMS) and the actual selection within the mode.

Autopilot Bug

The magenta triangle shows the selected course for the autopilot in HDG mode.

Data Presented

On the Left Hand Side

- The navigation mode: If VOR is selected and it is an ILS station, then ILS is displayed as soon as the glidescope information is in range.
- ...N: Distance to destination in nautical miles. If there are only dots and no number, then the station is not yet in range for distance calculation. The distance for ILS is displayed for the glidescope when available, else as VOR.
- ...M: Time to destination in minutes and seconds given current speed (or estimated with 400 kt if the speed is below 100 kt). The calculation depends on the availability of the distance.
- The name of the VOR/ILS station if available (e.g. EHE). Typically, ILS stations start with an “I” in the name.
- The TACAN channel (e.g. 029X)

The data points belonging to the currently selected mode (e.g. VOR vs. TACAN) are shown in cyan colour, otherwise white.

On the Right Hand Side

If there is an active flightplan (in green):

- The number of the current waypoint
- The distance to the current waypoint

A bit under the middle (in magenta): The selected altitude hold for the autopilot ALT mode.

At the bottom (in blue): the actual wind direction and wind speed. Please be aware that the indicated wind direction is relative to the aircraft (like the compass rose). The arrow point towards where the wind is coming from.

5.4.2 Electronic Attitude Direction Indicator (EADI)



The Sphere

The sphere in the middle shows attitude data (pitch and roll). If the nose is vertical up, then the sphere will be totally blue. If the nose is vertical down, then the sphere will be totally brown (you are flying directly into ground).

ILS Guides

If the navigation mode is either `NAV1` or `NAV2` and the flight mode is `Approach`, then glideslope an lateral guidance lines are shown if the station provides ILS and the following is true:

- Heading data is in range
- Glidslope data is in range
- The heading deviation is less than 90 degrees.

You change the navigation mode in the `EHSI` page. If the mode is not `NAV1` or `NAV2`, then on the top right `NAV*` is displayed and you can change to `NAV1` using the button above. Otherwise you can toggle between `NAV1` and `NAV2` with the button above.

Data Presented

On the top of the screen HUD data is repeated: speed (left side), heading (middle), altitude (right side).

In the middle of the screen navigation data is presented - left side is visible only if navigation mode is `DATA`, right side is for waypoints:

- the selected number
- the time in minutes (e.g. `023/`) given current speed (or estimated with 400 kt if the speed is below 100 kt)
- the distance (e.g. `32N`).

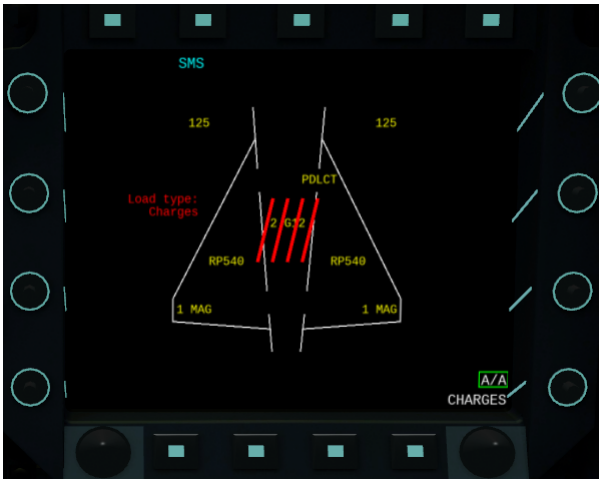
5.4.3 Store Management system (SMS)

The SMS page shows the currently loaded weapons incl. external tanks. A few things to note on the picture below:

- The red text on the left indicates that the load type for the Flight Control System should be changed given the heavy load of bombs. That can be done with the toggle at the bottom left.
- A gun is loaded (CC422 gun pod on a M2000D) and therefore at the top to the left you can see the remaining bullet count. If the page would be displayed in a M2000-5, then “CAN” would be displayed each of the 2 cannons.
- The currently selected store will have a yellow rectangle border. You can only select weapons (using `Key: w`) and only the first available weapon of the same type.
- The red stripes in the middle indicate that there is weight on wheels.

- At the top of the page below the middle button you can read “PPA” - which will be the next page displayed.

The abbreviations used for the stores are available in the weapons overview table in Section 11.1 (a number in front of the abbreviations means the number of this weapon at the station).



5.4.4 Poste de Préparation Armement (PPA)

The PPA is a weapons configuration panel. In the middle of the screen it shows the selected weapon plus the remaining number of this type.

At the top of the page a reminder for the pilot is displayed: “Damage: Off” means that the OPRF damage has not been enabled and therefore weapons will not generate damage when they hit something.

The displayed menus depend on the chosen weapon and sometimes on previously chosen menu items (e.g. the ripple distance is only shown, if ripple mode is set to more than 1).



5.4.5 Configuration Screen (CFG)

This MFD page allows to dynamically configure different areas of the aircraft before and during flight:

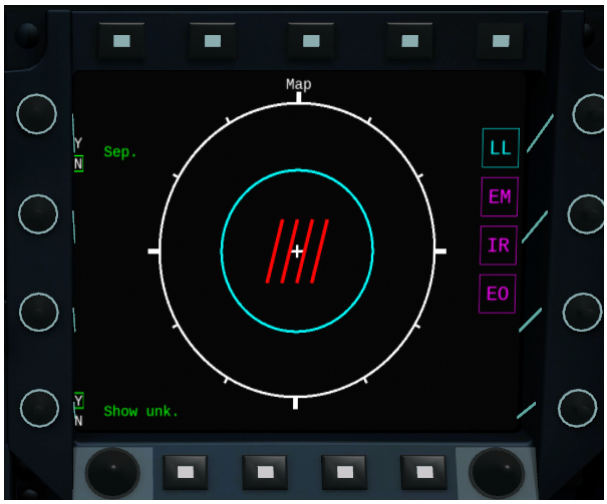
- Bingo fuel: between 100 and 3000 kg. Button up increases by 100 kg, button down decreases by 10 kg.
- Selected altitude (*alt. aff.* = altitude affiché for the autopilot): between 1000 and 50’000 ft. Button up increases by 1000 ft, button down decreases by 100 ft.

NB: this page is based on pure guess how it might work in the real thing and what might be configurable.

5.4.6 Radar Warning Receiver (RWR) and Counter-Measures

The radar warning receiver (RWR⇗) screen is a combination of a RWR display and a counter-measures dispenser

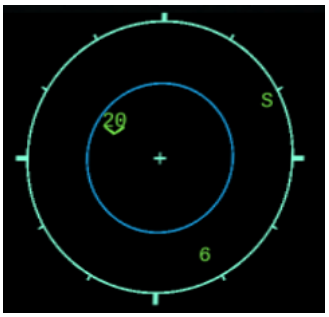
display.



On the left side of the screen there are 2 menu items for the RWR:

- Separation: whether the symbols should be dispersed a bit to make them more readable (but this changes the relative bearing).
- Unknown: whether to show radar sources, which cannot be interpreted.

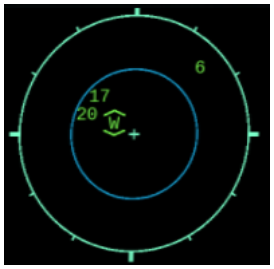
RWR



The RWR displays a maximum of 12 threats. High level threats (e.g. with an STT lock or actively guiding a missile) are displayed within the blue centre ring. Lower level threats are displayed closer to the outer ring. I.e. the distance from centre is an interpretation of threat and not a real distance. The position is a top-down view around your aircraft (own aircraft nose towards up/North).

NB: most signals from OPRF are only available, if MP Damage is on!

Different types of threats are displayed with different symbols according to USA/NATO standards (i.e. not according to French symbology at the moment). `U` is for unknown threat, `S` is for surveillance aircraft (e.g. AWACS⇗ - which typically cannot shoot), and `AI` is for aircraft which have not yet been classified in OPRF.



If there is a chevron below the symbol, then the threat has a radar lock on you. If there is a hat on top of the symbol, then the threat is the either source of an active missile or guiding a semi-active missile.

Only one missile in the air can be displayed - even though several might be in the air at the same time. The missile is shown with the symbol `W` close to the centre - again the distance is not the real distance and only the bearing relative to your aircraft is shown. If a missile is in the air, then the related threats are blinking once per second.

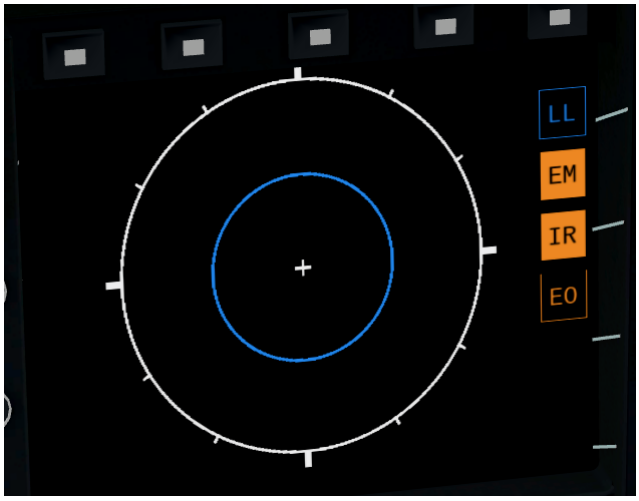
In addition to the visual indications there are sounds (refreshed every 0.5 seconds):

- A new threat has been detected: continuous 1 kHz tone for 0.5 seconds.
- A new radar lock (STT) has been detected: 1 kHz tone chopped at 25Hz for 0.5 seconds.
- A semi-active missile is being supported: 1 kHz tone chopped at 25Hz for 0.5 seconds repeating after 0.5 seconds of silence.
- An active radar missile is in the air: continuous 1 kHz tone chopped at 25Hz until the missile is not detected any more.

Sounds can be heard even if the RWR is not shown on any screen.

Counter-Measures Dispenser Display

At the right side of the RWR there are 4 indicators for dispensed counter-measures (flares and chaff). It is a "could-be" interpretation of the decoy dispenser lights on the right top of the M2000-C canopy.



- `LL` = decoy dispenser (Lance-Leurres) - blue: blinks when counter-measures are being dispensed.
- `EM` = chaff (Électro-magnétique) - amber: blinks when remaining quantity is at or below 20. Steady light when remaining quantity is at 0 (empty).
- `IR` = flares (Infrarouges) - amber: blinks when remaining quantity is at or below 20. Steady light when remaining quantity is at 0 (empty).
- `EO` = electro-optical (Électro-optique) - amber: not simulated.

The total quantity of counter-measures simulated is 120. 2 are dispensed every second. No difference is made between flares and chaff in the simulation. Use `Key: q` to start dispensing and `Key: q` to stop dispensing.

5.4.7 Map

The map page is a temporary replacement for a real implementation in the VTB. Using the lower right toggle you can zoom and and out of the map.

The map is based OpenStreetMap⇗ and shows only the position of one's own aircraft in the middle.

Depending on the network connection it might take a while for parts of the map (tiles) to load. Once loaded the tiles get cached and should therefore be available further on.



6 Flying

6.1 Basic Operations

6.1.1 Start-up

When your aircraft is cold and dark before start-up, the external generator and fuel truck are connected to your aircraft, and protections (in red) will be on your aircraft, as well as chocks. They will disappear automatically as you start up.

You can either auto-start your aircraft with `Key: }` or press button `autostart` in the dialogue opened by using menu `Mirage 2000 -> Configuration`.

6.1.2 Taxi

The Mirage can taxi at relatively high speeds, usually below 40kts, but you shouldn't go faster than 20kts when turning. Even if the track width is larger than usual, you should always stay at safe speeds.

The front gear can turn 90° to the left and to the right, so a turn radius of 3 meters is reachable. You might need to add thrust, as it is difficult to move with the front gear almost entirely turned. Do not push on the rudder pedals too much when braking, otherwise the aircraft may fall on one side or another if you turn too tight.

The taxi lights have an angle of about 30 degrees in front of the aircraft and are not attached to the moving part of the front gear, so they will always light in front.

NB: The nose wheel steering gets automatically disabled above 40 kt and cannot be enabled as long as ground speed is faster than 40 kt. The nose wheel steering can be toggled/re-enabled with `Key: S`.

6.1.3 Take-off

Line up on the best runway for the current weather conditions and ask for clearance. Once the clearance is given and after releasing the parking brake, spool up while keeping the pedal brakes enabled. Then, throttle up to 99% RPM (military power) for long runways, 100%+ (with afterburners) for short runways or with heavy loadout. Rotation speed is at 120 knots when using a clean loadout, 140 knots if with payload. After this, bring the inverted "T" in the HUD on the line of horizon and you should be able to lift off at 170-190 knots, depending on your loadout. Do not exceed 13° nose angle or you will tail-strike. Even though the Mirage is airborne, you might feel that the plane with high nose angle "floats" over the runway until reaching well beyond 200 knots. Raise the gears before reaching 260 knots with `Key: g` or by clicking the red gear lever.

The Mirage needs long runway distances to take off, as it hasn't got a separate horizontal tail stabilizer/elevator. Because of the elevons' position (trailing edge of the wing), they create a considerable loss of lift when they are moved up to rotate, so higher rotation speeds are needed. Also, the delta wing is often called "flying airbrake", as they generate lots of drag at high angles of attack, like at take-off. This is why high speeds and long runways are needed.

6.1.4 Initial Climb

After taking off, retract the landing gear before reaching 280kts (maximum gear extension speed). You are advised to follow the runway heading until you are at an acceptable speed (approximatively 200 knots), but if you used reheat and you are not heavy, you can turn as soon as you are at 200 knots. However, you have to watch your speed so you don't stall - at this altitude, a stall doesn't forgive.

After having chosen the correct heading and having attained 250 knots, you can commence your climb with full military power, at the beginning around 2,000 feet per minute to gain speed, and when you have reached your optimal climb speed you may pull the nose up to climb at a rate of 4,000 feet per minute. Remember that if you are faster, you will have an lower angle of attack and will thus drag less, and your elevons will not need to be pulled fully up (and thus will not create too much additional loss of lift), so it is advised to speed up before climbing too fast.

6.1.5 Climb

The Mirage can climb exceedingly fast, with a maximum climb rate of 6,000 feet per minute. With full tanks and air-policing⇗ loadout, it is able to climb with full military power at a rate between 4,000 feet and 6,000 feet without bleeding speed (between 250 and 300 knots). With full afterburners and in the same conditions, the climb rate can go up to 12,000 feet, still in zero/zero conditions. If going vertical with full afterburners and a base speed of 400 knots, you can reach 20,000 feet before having to push the nose back down. This is the fastest way of climbing.

6.1.6 Cruise

The Mirage 2000 normally cruises between angels 35 and angels 40 (35,000 - 40,000 feet), and this is the best altitude for Mach 2+ flights. It can also go above 50,000 feet (the pilot would need special equipment - not simulated yet), and up to 80,000 feet, but with difficulties. The optimal cruise altitude is 36,000 feet.

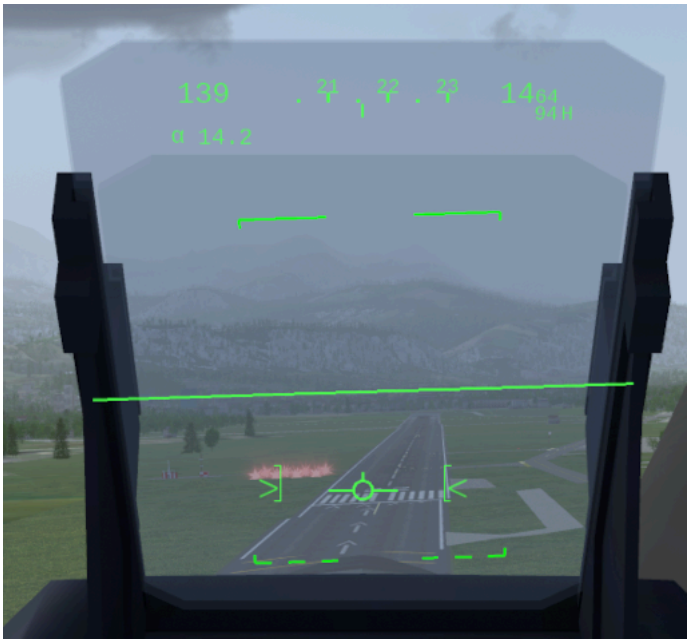
While it has a bi-sonic flight possibility, it consumes lots of fuel. It is thus advised to stay sub-sonic during cruise flights, without using afterburners. With cargo loadout and without using afterburners, the Mirage can fly for more than two hours. For long cruises, it is advised to use the navigation, altitude and speed autopilots.

6.1.7 Visual Landing

Make sure you have plenty of height and distance to align with the runway and land. Change the flight mode to `Approach` (cf. Section 6.2).

Lower the gears when you are below 250 knots, then approach the runway at 175 knots (for 15% fuel left; add 5 knots for every 20% additional fuel - and a bit more if you bring weapons home). Put the landing lights and navigation lights on. On final approach, hold an angle of attack of ca. 14 degrees. When passing the runway threshold, do a short flare and hold angle of attack at ca. 14 degrees. You should land at 145 knots (for 15% fuel left; add 5 knots every 20% additional fuel).

The HUD helps with visual landing. Once aligned with the runway, put the flight path vector on the runway threshold, make sure that the flight path vector is between the 2 approach brackets in the lower part of the HUD (indication for angle of attack (AoA)⇗ is at degrees) and use the throttle to keep the AoA as well as the speed in the right regime:



The height of the approach bracket is 1° (+/- 0.5 deg from 14).

When touching down and the 2 main wheels are on ground, the HUD automatically changes to flight mode `Ground`. Keep the nose up at around 13 degrees to do aerodynamic braking (cf. image below - inverted "T" cue in the HUD is a bit high and should be on the line of horizon). Wait for the aircraft to slow down to at least 135 knots, bring the nose-wheel down and first then apply pedal brakes. Use the brake parachute (`Key: o` to deploy) if necessary (heavy loadout or short runway) - and then release when not needed any more (same `Key: o`). Aerodynamic breaking using a high angle of attack when landed is normal procedure. However, do not tilt the aircraft more than 13 degrees due to the possibility of a tail strake. NB: it is by procedure forbidden to use aerodynamic braking below 100 kts - even though it might be possible: you have to get weight on the nose wheel. The reason for this required procedure is that depending on speed, configuration, weight distribution (e.g. fuel) etc. it might suddenly not be possible anymore to bring the nose down given loss of effectiveness of the elevons vs. drag forces vs. weight forces.



Note that nose-wheel steering only be engaged below 40 kts.

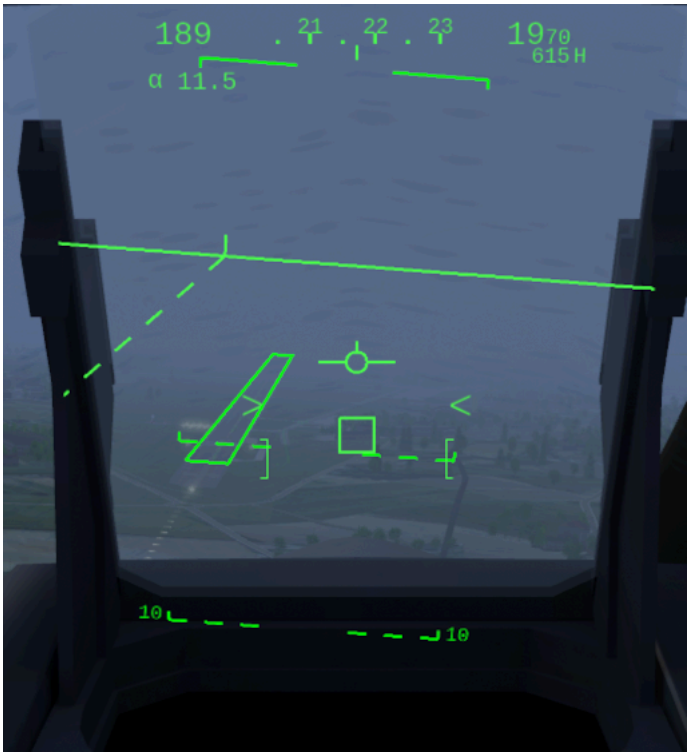
6.1.8 Landing with ILS

In bad weather or at night the Instrument Landing System (ILS) can help to find the runway. The HUD of the mirage displays cues from the ILS as shown in the next image:

- The synthetic runway.
- The stippled line to the left shows the runway axis direction. If it is not on the synthetic runway symbol, then you

have to fly towards the side of the stippled line to align with the runway.

- A guidance window (square). To land correctly the airplane must be steered such that the flight path vector overlaps the guidance window - while also being correct in relation to the approach brackets (see chapter above).



A few things must be true for the symbology to be displayed:

- The ILS frequency must be set through menu `Equipment -> Radio Frequencies` to the runway (you can get the frequency from dialogue `Location -> Select an Airport`).
- The radial runway degrees must be set through menu `Equipment -> Radio Frequencies` - you can get the value from dialogue `Location -> Select an Airport`.
- The radio channel must be selected.
- The runway axis and the guidance window are shown as soon as the localizer has been caught.
- The runway is only shown if both the localizer has been caught and within the glide slope (max ca. 7 degrees lateral deviation from the runway direction) and max 10 nm distance from the runway.

6.2 Flight Modes

There are 4 flight modes:

- `Ground`: Default. Used for take-off and during aerobraking when landing.
- `Approach`: Used during landing before touch down.
- `Navigation`: As the name says - in the air
- `Attack`: when a weapon is selected for air/air and air/ground attack.

NB: the real Mirage has mode modes. E.g. there are several sub-modes to navigation, which are not yet simulated.

The main difference between the flight modes are:

- The view position and zoom.
- What is displayed on the HUD (Section 5.1).

If there is weight on the main wheels, then flight mode `Ground` is automatically selected. Otherwise the possibilities to change between the different modes is as presented in the table below (left columns: from mode; top headers: to mode):

From / to	Navigation	Approach	Attack
Ground	Key: g (gear up)	n/a	n/a
Navigation	n/a	Key: h	Key: w
Approach	Key: h	n/a	n/a
Attack	Key: h	n/a	n/a

If you press the Key: w and no weapon can be selected, then mode Navigation is automatically selected.

6.3 In-Cockpit Views

In order to see the MFDs and the VTB better as well as seeing better the gauges on each side of the VTM, you can use key bindings:

- Key: n: view the left MFD and the VTB
- Key: N: view the right MDF and the VTB
- Key: c: view the VTM
- Key: ctrl-n: reset the view to the position of the current flight mode

Depending on the flight mode you need to adjust the seat to see better - including the symbology in the HUD. At take-off you should seat raised to see the symbology in the HUD. During approach/landing and ground attack your seat should be adjusted upwards, so you can see the point of the nose, etc.

- Key: shift-PageUp: adjust the seat up
- Key: shift-PageDown: adjust the seat down

It is highly recommended to use a head-tracking device. There are FlightGear Wikipedia articles for both FaceTrackNoIR and Opentrack. If you are using the Headtracker AddOn:

- changing the in-cockpit views will automatically toggle the headtracker
- use Key: ctrl+h to manually toggle the view (instead of using the menu)
- use Key: ctrl+t to manually toggle the translation (instead of using the menu). If headtracking is on and translation is on, then you can better see the indicators to the left and right of the HUD, but the seat will be default. To use the seat up/down changes, you have to disable translation.

6.4 Other Stuff

6.4.1 True vs. Magnetic North

To switch between displaying heading in true North vs. magnetic North, use the second button from right on the lower button row of the VTB.

6.4.2 Nose Wheel Steering

Use Key: S to toggle nose wheel steering.

Nose wheel steering is automatically disabled above 40 kt and cannot be enabled as long as ground speed is faster than 40 kt.

The steering degree is depending on the ground speed: at 1 kt it is 45° down to 10° at 30 kt.

NB: Nose Wheel Steering is “Dirigeabilité Roue Avant” in French, which is abbreviated “DIRAV” - and therefore the name on the blueish control light on the gears panel.

6.4.3 Miscellaneous

- When heavily loaded, the fly-by-wire system (FBW) should be set correctly to lower allowable flight limits in order to save the structure. Use the SMS page (Section 5.4.3).

- Afterburners engage when throttle moves past 85% in real-life, 90% in-sim.
- The 1700l and 2200l droppable tanks are not supersonic, but the 1300l is.

7 The Autopilot system

The Mirage has a rather simple (but not simplistic) autopilot, which can be either `off`, `on` in a specific mode or in `standby`. The following modes are available:

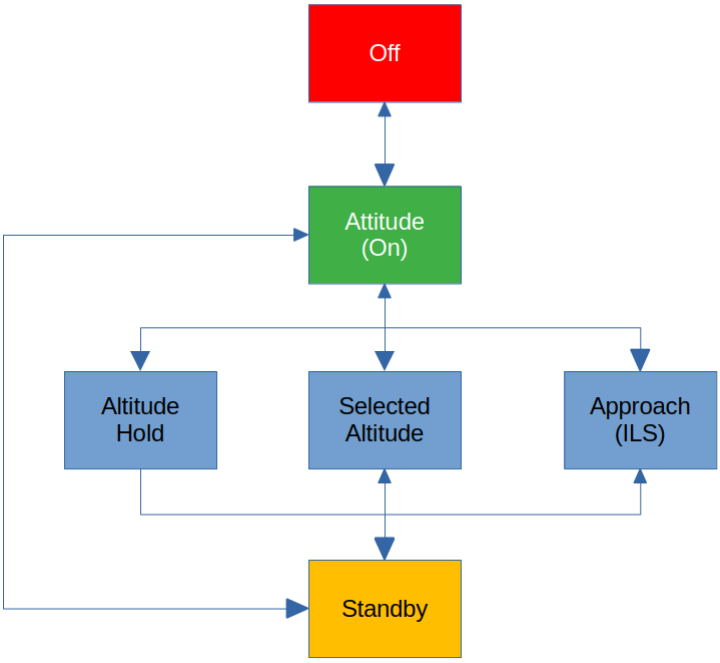
- Attitude hold (`PA` - basic mode): keep the pitch and roll attitude of the aircraft as maintained or indicated
- Altitude hold (`ALT` - advanced mode): maintain the altitude as captured at the moment when the `ALT` button was pressed.
- Selected altitude hold (`ALT AFF` - advanced mode): capture and then maintain the altitude as pre-selected in Section 5.4.5.
- (mode not in use)
- Automatic approach (`LG` - advanced mode): capture / maintain heading and pitch based ILS signal as well as pre-selected runway heading (*CP - cap vrai piste*) and glideslope (*PD – pente désirée*)

Please be aware:

- There is no autopilot mode for speed. In all modes the pilot has to actively use the throttle to maintain the desired speed.
- There is no autopilot mode to follow waypoints.
- The *autopilot* and *terrain following* are not the same thing. And currently terrain following is not implemented.
- The default FlightGear autopilot and autopilot dialogue are not available.

To change modes do the following:

- To engage or disengage the autopilot press either the `PA` button or use `Key: BACKSPACE`. If the autopilot is in one of the three advanced modes, then you have to toggle twice to disengage (once to get from an advanced mode to the basic mode and then to disengage altogether).
- To engage or disengage standby mode use `Key: ctrl + a` (there is no button on the panel). Does only work if the autopilot is in a basic or advanced mode.
- To engage one of the three advanced modes (`ALT`, `ALT AFF`, `LG`) press the respective buttons on the panel (there is no key binding). To disengage one of the three advanced modes either choose a different advanced mode or press the `PA` button to get into basic mode.



The following features of the original autopilot are not yet implemented:

- Moving the stick more than 50% will not disengage the autopilot.
- The autopilot does not wait to be fully engaged until the stick is in the neutral position.
- When the autopilot is engaged, you cannot use the trim buttons to change pitch attitude and bearing.
- Some specific yellow buttons will not blink in specific situations - instead they will illuminate steadily.

8 The Radar System

8.1 The Real Radar RDY

The -5 version of the Mirage 2000 is equipped with the RDY (Radar Doppler Modèle Y - Radar Doppler Multitarget), which can be used in air-to-air and air-to-ground tasks. See descriptions in the English Wikipedia and radartutorial.eu.

8.2 The Modelled Radar

The modelled radar is a copy of the FlightGear F-16 Flight Control Radar, because it is already nicely modelled and the F-16 radar most probably is quite similar to the RDY-versions.

Note In reality the different versions of the Mirage 2000 have different radars and different radar modes, but this is not modelled.

8.2.1 Radar Range

The radar range can be increased by using `Key: R` - and decreased by using `Key: E`. Each key press is a factor 2 (e.g. from 20 nm to 40 nm)

For air-to-air combat the range is up to 160 nm (depending on the mode), for ground and sea modes it is up to 80 nm (for ground auto 40 nm).

When the radar mode changes (either by actively changing the radar mode or as a consequence of e.g. selecting a target), the range can change automatically.

8.2.2 Radar Modes

The radar has a set of modes and sub-modes (see the description in FlightGear F-16 Fire-Control Radar - bearing in mind that the displays are different in the M2000 and not all functionality is the same):

- **CRM**: Combined Radar Mode
  - RWS: Range while Search
  - TWS: Track while Search
  - LRS: Long Range Search
  - VS: Velocity Search
- **ACM**: Air combat Mode
  - ACM-20: 30 by 20 degrees HUD field of view
  - ACM-60: 10 by 60 degrees HUD field of view
  - ACM-BORE: boresight
- **SEA**: Sea Navigation
- **GM**: Ground Mapping
- **GMT**: Ground Moving Target

To change between main modes use `Key: Q` - to change between sub-modes use `Key: ctrl-q`.

8.2.3 Selecting Targets

When in TWS mode the next target can be selected using `Key: y` (can be used several times to cycle between targets). To deselect the current target use `Key: ctrl-y`.

In other modes use the arrows on your keyboard (`Key:` ⇐, ⇒, ⇑, ⇓) to move the cursor on the VTM. Use `Key: l` (lowercase L) to designate a target. You can also bind the cursor movement to your joystick of throttle like the following example (for vertical you just use the `cursor-slew-y property`):

```
<axis>
    <name>Hat Switch Front Horizontal</name>
    <desc>Slew cursor left/right</desc>
    <number>
        <windows>6</windows>
    </number>
    <binding>
        <command>property-scale</command>
        <property>controls/displays/cursor-slew-x</property>
        <power>1</power>
    </binding>
</axis>
```

8.2.4 Visibility of Other Targets When One Target Has Been Selected

For air-to-air: unless you are in Track While Scan (TWS) mode, you will only see the selected target and no other targets on the display. I.e. you are in `Single Target Track (STT)`, where the radar dedicates the whole radar beam to one contact and therefore looses track of the other potential targets.

For air-to-ground in the simulated M2000 the model is `Fixed Target Track (FTT)` for ground / sea and `Ground Moving Target Track (GM(T)T)`, which again dedicate the whole radar beam to the selected target. Therefore, once you have selected/locked one target, all other targets disappear from the radar screen.

When you un-lock/deselect a target (`Key: ctrl-y`), the other contacts will start to appear again.

8.2.5 Target symbols

- Air modes:
  - Contact: open rectangle with a line showing the direction the target is flying at (the longer the line, the faster)

- Selected/locked target: cross with line as above
- IFF friendly contact during interrogation (not yet implemented): filled diamond
- IFF foe contact during interrogation (not yet implemented): v-formed

- Ground modes:
  - Contact: filled circle without direction or speed information
  - Selected/locked target: cross
  - Spotted laser/GPS target: diamond without fill
  - designated laser/GPS target: diamond with dot in the middle

9 IFF

An Identification Friend or Foe system is for combat scenarios. Despite the name, IFF can only positively identify friendly aircraft, it cannot identify foes. An enemy aircraft is undistinguishable from e.g. a civilian aircraft, or an aircraft with a non-functioning IFF system.

IFF interrogations from elsewhere towards your aircraft is supported (i.e. others can tell whether you are a friend of foe). See Section 4.13 for a description of the cockpit instruments.

IFF interrogations from your aircraft towards other aircraft is not yet modelled (i.e. you cannot tell friends from foes).

The Mirage 2000 supports 2 IFF channels. According to [ref: D04] the IFF code gets automatically switched at midnight from A to B. This is not modelled.

The IFF channel codes are set in menu item `Mission Preplanning` accessed through menu `Mirage 2000` (if damage is on, then it can only be done on ground!).

10 Link 16

Link 16 (aka. data link) is not implemented.

11 Weapons and Armament

11.1 Overview of Available Weapons in the FlightGear version

The abbreviation column refers to the abbreviations used in the SMS page (see Section 5.4.3).

Abbrev	Weapon
CAN	30mm Cannon (internal) with DEFA 554
GUN	CC422 (gun pod) with DEFA 553
MAG	Matra R550 Magic 2
SUP	Matra Super 530D
IR	MICA IR
EM	MICA EM
AM39	AM39 Exocet
AS37A	AS-37 Armat (Martel)
AS30L	AS-30L
APACH	APACHE
SCALP	SCALP-EG
ASMP	ASMP-A
G12	GBU-12
G24	GBU-24

BL	SAMP Mk-82 (FR: bombe lisse)
BF	SAMP Mk-82 Snake Eye (FR: bombe freinée)
RP520	1300 l drop tank
RP540	2000 l drop tank
RP500	1700 l drop tank
PDLCT	PDLCT targeting pod (TGP)
SMOKE	Smoke Pod

The main source of information about available weapons as well as which variant can carry which weapons is ref[B10]:

- `C` variant plus general descriptions: pages 45-56
- `N` variant: pages 132-134
- `D` variant: pages 164-166 for pods, pages 174-175 for APACHE/SCALP, page 163 & 177 for weapons on stations
- `-5` variant: page 207.

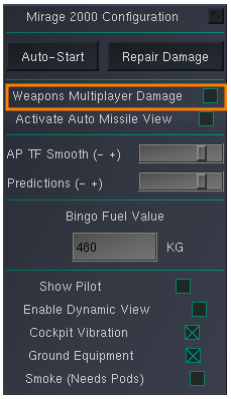
Just because a variant has the capability to use a weapon does not mean, that a given air force (incl. the French) has used them. E.g.:

- the Hellenic Air Force (HAF) uses only the SCALP and the AM39 Exocet for ground attack (ref[A08, A09])
- France does not seem to use the AM39 on Mirage 2000x - it uses other fighter jets
- Egypt was probably the only air force to use the ARMAT (ref[A10])
- Some weapons like the BLG 66 Belouga have been withdrawn from service by e.g. France due to political concerns regarding the use of cluster ammunition.

11.2 MP Damage

You can only damage other aircraft or MP enabled assets if (a) you have MP damage on and (b) MP damage is on in the target!

To enable or disable MP damage you have to be on the ground and use the configuration through menu item `Mirage 2000 -> Configuration`.

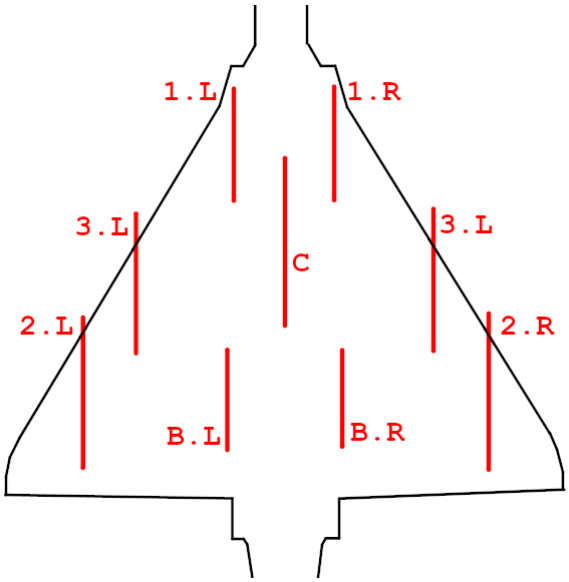


When damage is on, some features are enabled/disabled automatically to make aerial combat more realistic:

- Black-out/red-out is enabled
- Simulation rate is set to normal
- Emergency altitude increase is disabled
- Fuel and payload dialogue will not be available in air
- Mission Preplanning dialogue will not be available in air
- Map traffic is disabled
- MP pilot list is disabled

11.3 Adding/Removing Weapons

There are 9 weapon stations. The numbering of weapon stations is as follows:



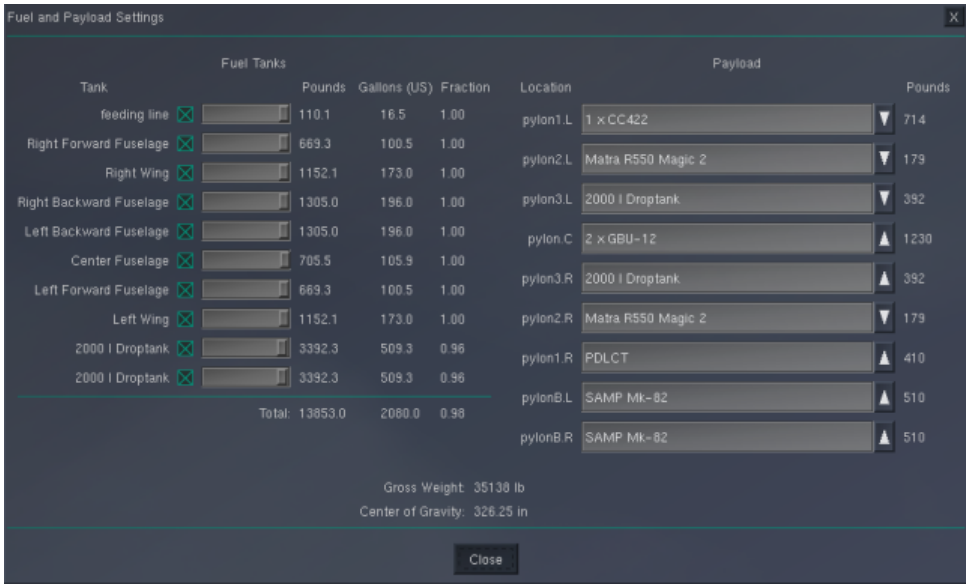
Depending on the aircraft version and the weapon station, possible weapon systems or external fuel tanks may vary. The available payload for each weapon station is as follows:

Payload	2.L	3.L	B.L	1.L	C	1.R	B.R	3.R	2.R
RPL-541/542 (2000L fuel tank)		5BD						5BD	
RPL-501/502 (1700L fuel tank)		5BD						5BD	
RPL-522 (1300L fuel tank)					5BD				
Matra Super 530D			5				5		
Matra R550 Magic II	5BD								5BD
MBDA MICA IR	5B								5B
MBDA MICA EM			5B	5B		5B	5B		
AS30 Laser		D						D	
AS37 Armat		5						5	
AM39 Exocet		5BD						5BD	
SCALP					5BD				
APACHE					5BD				
GBU-12 Paveway II			D	D		D	D		
2x GBU-12 Paveway II					5BD				
GBU-24					5BD				
SAMP Mk-82			5BD	5BD		5BD	5BD		
SAMP Mk-82SE			5BD	5BD		5BD	5BD		
2x SAMP Mk-82		5BD			5BD			5BD	
2x SAMP Mk-82SE		5BD			5BD			5BD	
PDLCT (targeting pod)						5BD			
CC422 (gun pod)				D*					
Smoke pod (white)	5BD								5BD
ASMP-A					N				

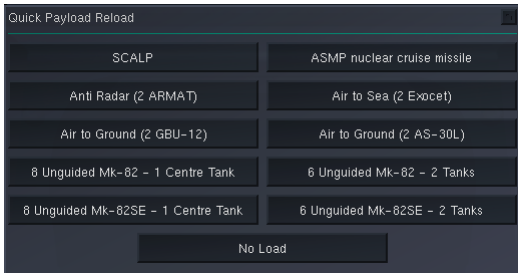
Key: 5 refers to the 2000-5, B to the 2000-5B, D to the 2000D/N, D to only the 2000D and N to only the 2000N.*

Apart from the available weapons per station, the simulation does currently not impose additional restrictions w.r.t. the combination of weapons. Pictures of real aircraft show that quite some variance is in action. However, it of course makes sense to e.g. have a pod installed if you need laser guidance - or that the weights on both sides of the fuselage

are in balance. And the real Mirage can only carry one type of ground attack weapon at a time (e.g. not SAMP Mk-82 and SAMP MK-82SE at the same time).



To make loading weapons easier, a set of typical loads for a given variant is available in menu item `Mirage 2000 -> Payload Selection`. Using either of them will also replenish the DEFA cannon as well as the onboard chaffs and flares.



NB: you cannot change your payload when MP Damage is on and your aircraft is moving!

11.4 Basic Keys

- `Key:` `m`: toggle master arm
- `Key:` `w`: cycle through weapons - also used to get into flight mode `Attack`
- `Key:` `e`: weapons trigger for guns, missiles and bombs
- `Key:` `M`: cycle through weapon guidance modes (e.g. LDP for laser guided ammunition)
- `Key:` `y`: select next target in TWS mode
- `Key:` `l` (lowercase L): designate target under the cursor
- `Key:` `ctrl-l` (lowercase L): Fast snipe and designate clicked target for laser
- `Key:` `ctrl-y`: deselect the current target use (air and ground modes)

11.5 Configuring the Active Weapon

The stores management system (see Section 5.4.3) shows the available and active weapons. Depending on the active weapon some configurations can be done through the weapons configuration panel (see Section 5.4.4).

11.6 Shooting

Make sure that the master arm is on. The HUD displays the diverse reticle in `Attack` mode also if master arm is off. If the master arm is off, then the weapon mode on the left side of the HUD is flashing.

Use the trigger (`Key: e`) for releasing a weapon no matter the type.

11.7 Smoke Pods

You can use smoke pods on the left outer station (2.L) and/or the right outer station (2.R). Use `Key: w` to switch to smoke pods and then use the weapons configuration panel (see Section 5.4.4) to toggle smoke on and off.

11.8 Ground Attack

11.8.1 Aiming Modes

CCIP

A Constantly Computed Impact Point (CCIP) is in the Mirage shown as a line from the flight path indicator (top) to the impact point (bottom). When the impact point is just above the target, then you release the bomb(s).

The release in the image below will not have a good accuracy, because the airplane is slightly banked to the right - for a good shot the airplane should be horizontal.



CCIP vs CCRP is automatically chosen depending on the weapon. CCIP is chosen for the `Snake eye` retarded bombs, but is not available for the standard Mk82 bomb as well as for guided bombs (e.g. GBU-12) - for those you need to use CCRP.

CCRP

The Constantly Computed Release Point (CCRP) requires a target to have been designated - either with the cursor in ground attack radar mode or using a laser or GPS coordinates.

The following picture shows CCRP guidance for a laser guided GBU12 bomb a few seconds before the ideal release point. The diamond at the top shows with its wings that the pilot should navigate the airplane a bit to the right. The short horizontal line below the diamond moves up from below the closer the release point is. On the right hand side the distance to the release point (not to the target) is shown.



The next picture shows a similar situation, but this time the weapon is a dumb free-fall bomb and the target has been designated using the ground radar.



Laser or GPS Designation

In the `Mirage 2000` menu there is a menu point `Ground Targeting`, which will display the following dialogue:



It is important to do all steps sequentially: (a) acquire a targets coordinates, (b) make it a spotted target, (c) designate the target for engagement.

You can acquire coordinates in two ways: either write the lon and lat directly into the fields - or look outside of the cockpit and click on the ground where you want to pick the coordinate from. Then press the top button (it will be disabled if no laser designation pod has been added to the airplane).

Always the primary coordinates will be used to create/update targets, but a secondary pair can be input as well. A button can swap the primary and secondary coordinates.

A spotted target (simulating what would be done with a laser) can be created with a button based on the primary coordinates. NB: the view can briefly changed automatically, such that the coordinate including its elevation above sea level can be fetched by FlightGear behind the scenes.

The spotted target can then become a designated target using another button.

Alternatively, using `Key: ctrl-l` you can fast spot and designate the clicked target for laser.

A FLIR view (`Key: F6`) can be activated and the button `Focus FLIR on Sniped Target` will then point the laser to the sniped coordinates. Thereby the coordinate can be improved by clicking exactly on the target through the FLIR and then updating the target.

NB: The laser-guided bombs and the AS-30L have a relatively narrow seeker field of view of ca. 30 degrees. Therefore, the aircraft nose must point to the designated target at launch time, such that the seeker can pick it up.

11.8.2 MK-82 / MK-82SE and GBU-12/GBU-24

On the PPA the following settings can be done:

- The number of bombs to ripple (min = 1, max = 18, 0 is not available). You should not ripple GBU's.
- The distance in metres between rippled bombs (min = 5, 10, 20, 30, …, max = 200).
- The fuze selector can be set to either `RET.` (retardé/delayed fuze), or `INST.` (instantaneous fuze) or `INERT.` (inertial fuze). However, this is not implemented and will always result in an instantaneous fuze.

NB: You cannot choose to release 2 bombs at once (dual mode). NB: The performance of your computer influences what the minimal distance between rippled bombs can be. If you have 60 fps and fly 600 kt then 5 metres is possible, but if you have 30 fps then 10 metres is the minimum distance, etc. If you fly 300 kt and 30 fps then 5 metres are possible, etc.

For the GBUs a laser target has to be designated (see Laser or GPS Designation), the weapons guidance mode must be `LDP` and airspeed at least 350 kt.

11.8.3 AS-30L(Laser)

The missile needs to be fired:

- below 32’800 ft
- below 5 g
- below mach 0.9
- below 45 degs of roll
- within 9.7 nm (18 km) of the target
- with a designated target and weapons guidance mode in `LDP`
- powered on
- the airplane nose must point within 16 degrees to the target, because the seeker field is only 32 degrees

For the designation of a laser target see Laser or GPS Designation. Once the missile has been launched, then the plane can turn away, because the target designation pod can illuminate by turning the laser pointer.

11.8.4 Cannons and Guns

The Mirage uses DEFA cannons:

- DEFA 554 for the single-seat Mirage 2000-5: the rate of fire can be changed to either `High` = 1800/min (0.033) or `Low` = 1200/min (0.05) using the PPA.
- DEFA 553 in the CC442 gun pod for the Mirage 2000D has a constant rate of fire of 1300/min (0.046).

In the PPA the gun mode can be set to either air-to-air or air-to-ground (“cannon air-sol”).

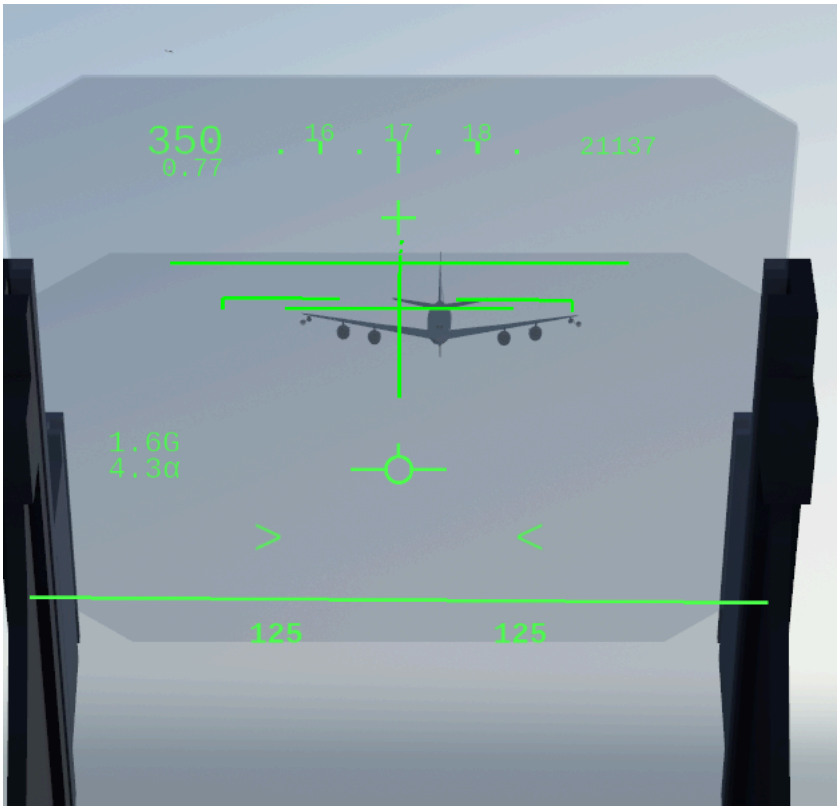


On the lower left of the PPA the symbology for shoot incitation in the HUD for air-to-air can be toggled between two

modes:

- CCLT (Calcul continu de la ligne de tir): tracer line (no radar needed)
- PRED (Prédictif): shoot incitation symbology is displayed if there is a radar lock and the geometry between the aircraft and the target allows it.

For air-to-air the wingspan of the expected target can be set between 7 and 40 metres using the switch on the lower right side (press up = +1 metre, press down = +5 metres). This influences the 600 m and 300 m markers on the tracer line in the HUD. In the picture below a plane with a wingspan over 40 metres is ca. 550 metres away and therefore a bit wider than the lower 600 m line and smaller than the 300 m line above.



11.8.5 AS-37 Armat (Anti-Radiation)

There is very little information available for the AS-37 Armat ("Anti Radiation MArTel") missile - especially how the aiming in the cockpit is done. And there is a lack of officially confirmed information abouts its use on Mirage 2000 variants as well as which countries might have been using it. Most probably requirements for not having to fit the launching aircraft with a lot of specialised equipment for the SEAD (Suppression of Enemy Air Defenses) can be fulfilled by the AS-37 (e.g. like stated in "RAAF's requirement under AIR 5398" in ref[A07]). Ref[A07] also states "With its high launch weight, heavyweight warhead and long range, the Armat is primarily an offensive strategic ARM designed to destroy Early Warning and Ground Control Intercept radars." Some sources indicate that the Armat had few (3) different seeker head options that could track only specific radar types. Because this is decided on ground there is limited possibility to chase opportunistic targets and cannot be used for (self) defence.

NB: if you have any information/hints available to make the implementation more realistic, then let the developers know.

Therefore, the implementation is purely fictional and takes inspiration from the DCS AV-8B Harrier AGM 122 Sidearm setup:

- Most probably the missile was only available in the -C version, maybe in the -D/N versions. This is why a rather primitive display system only using the HUD is used - only the -D and -5 versions have displays, which could show a page like in the F-16 or F-18 MFDs.
- The type of radar target has to be set before takeoff (to simulate that early versions of the AS-37 had 3 different seekers to be installed - even though the AS-37A most probably did not have different seekers): `GROUND` (GCI, radar towers - radar code `S` in OPRF), `SHIP` (frigates etc. - `SH`), `SAM` for larger static ground based SAMS (SA-3 `3`, S-200 `5`, S-300 `20`, MIM104D `P`), or `AAW` for smaller SAMS and self-propelled anti-aircraft warfare systems (Shilka `AA`, SA-6 `6` and Buk-M2 `17`) - this is really just a "random" choice for simulation.

- Power for the missile seeker must be explicitly activated.

For aiming you need to combine the RWR display (see Section 5.4.6) with information in the HUD:

- The HUD only displays a narrow field of view and filters on surface or naval assets which have an air radar. Unknown target types will not be displayed. A maximum of 8 threats are displayed (more severe threats are prioritised).
- The HUD is based on the passive radar homing detection capabilities of the missile, which has been chosen to be 50 nm (the max range of the missile is 54 nm) - which is much lower than the RWR capabilities.
- Therefore, you can use the RWR to spot potential targets and then steer the airplane into that direction until the target is displayed on the HUD (unless it is airborne or out of range). Remember that the distance from the centre in the RWR is not an indication of distance!

In the HUD all detected radiation emitters within the HUD field for view are displayed without any range information using the same type information as in the RWR. E.g. on the following screenshot you can see three radiation targets in the HUD - the same as shown in the RWR (where they overlap). The reticle is always in the centre and gets a double circle, when there is a lock. As with IR-missiles you will hear low and high volume sounds to indicate whether a lock has been acquired.



The missile needs to be fired:

- below 38’000 ft
- below 5 g
- below mach 0.9
- below 45 degs of roll
- within 50 nm of the target (more realistically below 35 nm if you launch it from pop-up)

Tips:

- Once a lock is acquired the reticle does not have to be kept directly over the target. However, if the reticle is moved too far away before the missile is shot, then the lock will get broken.
- The missile flies directly towards the target. If you launch too shallow, then it might hit the ground.
- After the first missile has been fired, the second missile’s power must be activated in the PPA.

11.8.6 AM39 Exocet (Anti Ship)

The missile needs to be fired:

- below 30’000 ft
- below 5 g
- below mach 0.9
- below 45 degs of roll
- within 38 nm of the target
- flying level and not below 250 ft (the missile motor starts frist after a drop time of 2 seconds and needs some time to accelerate the missile to stable flight). Flying level is really important, because if the nose points downwards, then in addition to the drop time the missile will accelerate towards the water without the pitch correction having a chance to react in due time.
- with a target selected in SEA radar mode

Tips:

- Power for the missile must be explicitly activated - also before you want to fire a second missile.

12 References

This page contains a structured selection of references, which can or have been used for modelling this FlightGear Mirage 2000.

12.1 Books

12.1.1 B01: Dassault Mirage: The Combat Log

Ref: Salvador Mafé Huertas, 1996; Schiffer Military History; ISBN: 0-7643-0168-3; 208 pages

Content:

- page 200: Picture of M2000-5 with all its weapons (picture by Dassault) - including amongst others the MATRA ARAMAT and rocket pods.
- page 205: Illustration of the (Mirage IIIE) Cyrano Ilbis radar air-to-ground sweep as well as the Cyrano Ilbis radar blind penetration mode.
- page 206: Illustration of Mirage IIIE level bombing and toss bombing.

12.1.2 B02: Dassault Mirage 2000: Flying in Air Forces around the World

Ref: Duke Hawkins, 2017?; HMH Publications; HMH-DH-003; ISBN: 978-2-9602488-2-1; 108 pages

Content:

- In general: High resolution and high detail photos of all parts for the Mirage - and all models.
- Very few cockpit pictures - and instruments are not turned on.
- Very few weapons pictures - and only a minor selection.

12.1.3 B03: 35 Years of Air Superiority

Ref: John Lekkas, 2023; Eagle Aviation; ISBN 978-618-85165-3-3; 124 pages

Content:

- In general: Tells the story (in Greek and English) of the Hellenic Air Force 331 squadron “Theseus” and their use of the Mirage 2000-5 Mk2.
- Quite a few high resolution cockpit and weapons pictures.
- Cockpit pictures: page 9, 52 (not turned on)
- Instrument pictures: page 25/81/88/89 (turned on - by night), page 80 (turned on - by night backseat), page88

(high resolution VTM)

- HUD pictures (guns against F-16): page 1/5/43/102

12.1.4 B04: Mirage 2000-5: Groupe de chasse 1/2 Cigognes

Ref: Alexandre Paringaux, 2017: Zéphyr Editions; “5”; ISBN 978-2-36118-249-6; 179 pages

Content:

- HUD pictures: page 7, 68 (simulator)
- VTB pictures: page 68 (simulator)
- VTM pictures: page 71 (plus all other instruments in high resolution - but probably simulator)
- Picture of leg tablet: page 66/67

12.1.5 B05: Mirage 2000D: Escadron de Chasse 3/3 Ardennes

Ref: Alexandre Paringaux, 2013: Zéphyr Editions; “1”; ISBN 978-2-36118-122-2; 80 pages

Content:

- Picture of leg tablet: page 57

12.1.6 B06: Dans le Repair du Mirage 2000D: Nancy-Ochey

Ref: Alexandre Paringaux, 2013; Zéphyr Editions; ISBN 978-2-36118-116-1; 144 pages

Content:

- A number of pictures of different weapon configurations - and some close-ups
- page 75: some instruments on the simulator consoles
- page 82: front cockpit by night with instruments running (in simulator)
- page 84: back cockpit by night with instruments running (in simulator)
- page 107: front cockpit of crashed plane

12.1.7 B07: Mirage 2000N: Escadron de Chasse 2/4 La Fayette

Ref: Alexandre Paringaux, 2016; Zéphyr Editions; “4”; ISBN 978-2-36118-211-3

Content:

- Many weapon configurations - also conventional weapons (page 94 & 96/97 4 GBU-12)
- page 111: night illuminated front cockpit with working VTB (in simulator)
- page 62: picture of leg tablet

12.1.8 B08: Les Forces Aériennes Stratégiques: 50 ans d’alerte nucléaire

Ref: Alexandre Paringaux, 2015; Zéphyr Editions; “3”; ISBN 978-2-36118-168-0

Content:

- page 4: picture of front cockpit 2000N by night illuminated (simulator)

12.1.9 B09: AMD-BA Mirage 2000D

Ref: Frédéric Lert; Les Matériels de l’Armée de l’Air et de l’Aéronaval; Histoire & Collections; ISBN 978-2-35250-253-1

12.1.10 B10: Mirage 2000: L’histoire and l’armée de l’Air de 1974 a nos jours

Ref: Hervé Beaumont, 2024; 2e édition; Sophia Editions; ISBN: 978-2-38514-081-6.

Content:

- Each version incl. the prototypes are described in depth in concise language - not so much “stories”.

12.2 Articles in Journals or BLOGS

12.2.1 Jets - L’aéronautique à réaction et son histoire

No 23, November 1997

A01: Riccardo Niccoli: En vol sur Mirage 2000-5 (4 pages)

- page 7: cockpit view of the front panel. Shows amongst others the 4 stripes indication of not working screens and the possibility to repeat the VTM radar screen on the VTB. And good view on the lower left panel with the 3*6 button table. The buttons are blue when illuminated, but not much.

12.2.2 World Air Power Journal

Volume 10 Autumn/Fall 1992

A02: Paul Jackson, Dassault Mirage 2000 (46 pages)

- Detailed description with lots of pictures of all current at that time variants and all user countries.
- Quite a few pictures of weapons - some of which are not used any more.
- page 60: instruments view of M-2000C and M-2000-5 (not very good resolution).

12.2.3 Nouvelle Revue d’Aeronautique et d’Astronautique

No. 2, 1997

A03: Georges Mas, Marc Pagliardini: Le Programme Radar RDY

Retrieved as “The RDY Radar Program” translation for requester NAIC/TAEA (National Air Intelligence Center), document control number: NAIC-98-0124, translation number: NAIC-ID(RD)T-0124-98 from internet.

12.2.4 Dassault Aviation - Engage!

Special Issue

A04: Mirage 2000-9

URL: https://www.dassault-aviation.com/wp-content/blogs.dir/1/files/2012/08/Mirage_2000-9_special_issue1.pdf

2 - 2003

A05: RDY2 multi role, multi function radar

A06: M53-P3 a new engine under the skin

12.2.5 Others

Air Power Australia

A07: Matra/BAe ALARM and Matra Armat

URL: https://www.ausairpower.net/alarm-armat.html

Key.Aero

A08: Multi-Role Hellenic Mirages (May 2017)

URL: https://www.key.aero/article/multi-role-hellenic-mirages

A09: Theseus over the Aegean (May 2020)

URL:

A10: To Kill a Radar: Analysing the now largely-forgotten ARMAT missile (Jan 2022)

URL: https://www.key.aero/article/kill-radar-analysing-now-largely-forgotten-armat-missile

12.3 Documents

12.3.1 D01: Fiche technique Mirage 2000D

URL: https://www.defense.gouv.fr/sites/default/files/air/Fiche%20technique%20Mirage%202000D.pdf

Dassault Mirage 2000D - Fiche Technique - Un Mirage Rénové

12.3.2 D02: Brétigny-sur-Orge - le A-2019-01-A

URL: https://www.defense.gouv.fr/sites/default/files/bea-e/A-2019-01-A.pdf

Bureau enquêtes accidents pour la sécurité de l'aéronautique d'État - Rapport d'enquête de sécurité - A-2019-01-A

Content - investigation of a Mirage 2000D crash during training at very low altitude on January 9th 2019.:

- Page 4: abbreviations
- Page 23: explanation and picture of the VTB visualisation for terrain following.
- Page 34, chapter 2.2.6.4: how the VTB and VTH react if flying below the set margin above terrain.
- Page 43, chapter 2.3.2.1: more text describing the terrain visualisation (which reads from right to left!).

12.3.3 D03: Mirage 2000-C - Manuel Pilote

Ca. 131 pages in French. Available as pdf-file on the internet.

12.3.4 D04: M2000-C User Manual DCS - RAZBAM

The user manual for the Mirage 2000-C simulated in the DCS combat simulator by RAZBAM. Available at RAZBAM DCS Manuals.

12.4 Links

- Armée de l'Air et de l'Éspace:
  - Home page: https://www.defense.gouv.fr/air
  - Mirage 2000D: https://www.defense.gouv.fr/air/nos-aeronefs/nos-avions/mirage-2000
  - Mirage 2000-5F: https://www.defense.gouv.fr/air/nos-aeronefs/nos-avions/mirage-2000-5f
  - Air Actualités: https://www.calameo.com/accounts/14334
  - YouTube channel: https://www.youtube.com/channel/UC9tdrNLs9QpsFZI050UARkQ
  - Escadron de Chasse 1/2 "Cigognes": https://www.gc1-2cigognes.fr/
  - Wikipedia FR: Liste des escadres AdlA
- Dassault:
  - Mirage 2000
- Others:
  - Forum thread: Mirage 2000 RDM/RDI/RDY Radars and other avionics
  - Blog post: quite a few details on systems and weapons with pictures
  - Thai military and Asian region blog post: a lot of details across variants - also weapons
  - Scalemates Dassault Mirage 2000 Walkarounds
  - Mirage 2000 discussion thread on air-defense.net (French)
  - Mirage 2000 discussion threads on DCS Forum

12.5 DVDs

12.5.1 V01: Mirage 2000 - 20 Years in Hellenic Air Force Service

Ref: www.11aviation.com; 87 minutes; Greek language / no subtitles

Content:

- File 05-3
  - 21:05 minutes: HUD on ground
- File 05-4
  - 2:33 min: HUD on ground/take-off
  - 5:13 - 07:50: (ditto)
  - 10:00: HUD landing
  - 11:00: HUD with seeker

12.5.2 V02: Mirage 2000-5

Ref: Hellenic Air Force Yearbook; Airutopia, David Maxwell; 2012 Special Projects; ca. 1 hour; mostly English with Greek subtitles.

Content:

- File 01_1:
  - 5:58: Roll-rate = 270 degs / second
  - 7:09: HUD
  - 27:45/28:21/20:03-29:24: target in HUD
- File 01_2:
  - 16:20: startup with engine gauge instrument
  - 18:00: weapons page display
  - 19:14: attitude page display
  - 19:21: VTB tactical display
  - 19:30: Navigation page display
- File 01_3:
  - 10:01 / 11:04: Backseater HUD

12.6 YouTube Videos

12.6.1 Y01: Mirage 2000 Planète Science

URL: https://www.youtube.com/watch?v=MoVsmaIs_pU

Content:

- At around 9 minutes for ca. 2 minutes: live images of the 5 screens

12.6.2 Y02: DGA : Le Mirage 2000B qui se prenait pour un Rafale

URL: https://www.youtube.com/watch?v=a5ZwYXixT28

Content:

- At around 6 minutes for ca. 1 minutes (and also before and after) close-up pictures of the CC422 gun-pod

12.6.3 Y03: Le Mirage 2000 - Documentaire sur l'aviation

URL: https://www.youtube.com/watch?v=jDCxWEynbu4

Content:

- At around 21:30 for ca. 15 seconds live pictures left MFD (in simulator)

12.6.4 Y04: AB Moteurs Mirage 2000

URL: https://www.youtube.com/watch?v=bFHF9j_LvPk

Content:

- From around 29-41 minutes (with interruptions): walk-through of the cockpit, which gives a good insight into the French terms used.
- Some live footage of HUD here and there.

12.6.5 Y05: Les Guerriers du Ciel - Mirage 2000, Le Chasseur Polyvalent (Documentaire)

URL: https://www.youtube.com/watch?v=VkRisFae93Y (same as https://www.youtube.com/watch?v=YjI7Tg75MD8)

Content:

- Presentation of the -C, -5 and N versions.
- At around 11:00: Close up view of the antennas / sensors on the tail and the back of a Mirage 2000D.
- At ca. 21:00: A few seconds of Mirage 2000-5 left side of front cockpit panel plus HUD sequence of missile launch (probably simulator)
- At around 31:00 - 32:30: Mirage 2000N simulator cockpit views

12.6.6 Y06: Mirage 2000 D : un appareil CRUCIAL avant le "tout-RAFALE"

URL: https://www.youtube.com/watch?v=uVWe9SUecsw

Content:

- Rénovation mi-vie (RMV) incl. description of the different pods (ATLIS, PDL-CTS, DAMOCLES, TALIOS) and the related guided weapons
- Around 11:03: photo of 2 loads for RM, one of them having a CC422 pod, 2 GBU and two MK82.
- At 15:15: backseat cockpit with new big screen

12.6.7 Y07: L'HISTOIRE des FANTASTIQUES MIRAGE de Dassault - doc complet (III, 5, 50, G, IV, F1, 2000)

URL: https://www.youtube.com/watch?v=GSDRy0jYgA0

Content:

- From around 48:01 until 1:07:10: M2000
- At around 1:05:01: HUD in weapons mode from Greek/Turkish M2000 vs. F-16 fight.

12.6.8 Y08: Mirage 2000 Cockpit - Documentaire avion de chasse Dassault

URL: https://www.youtube.com/watch?v=zqENHPav5P4

Content:

- Commented walk-through of the cockpit (displays black) of a M-2000C - close up

12.6.9 Y09: Mirage 2000 pour l' Ukraine? Mieux comprendre les rumeurs

URL: https://www.youtube.com/watch?v=GrNu8qAJ16s

12.6.10 Y10: A Mirage 2000 fires its cannon! A first

URL: https://www.youtube.com/watch?v=UB4Dl1hhHUc

Content:

- 04:48: Overview of the firing range with different targets
- 09:03: VTB image with and without LINK16

12.6.11 Y11: Turkish F-16 vs Hellenic Mirage 2000 Dogfight Video Analysis

URL: https://www.youtube.com/watch?v=0GGNbgV4crk

Content:

- Analysis of air-2-air combat with live HUD camera

12.6.12 Y12: French Mirage 2000 Landing at Berlin Schönefeld Airport for ILA 2012 HD

URL: https://www.youtube.com/watch?v=JcWj9M4LYDM&t=82s

Content:

- Aero braking by a M2000D/N

12.6.13 Y13: Mirage 2000-N BAN Hyères - Atterrissage

URL: https://www.youtube.com/watch?v=PQSUEYkh4yA

Content:

- Aero braking with air brakes on touch down

12.6.14 Y14: Hellenic mirage 2000 is out for hunt

URL: https://www.youtube.com/watch?v=uLpjhl8Yh3M

Content:

- Almost 4 minutes of HUD footage with diffreent A2A weapons in dogfight

12.6.15 Y15: Aviation - Crash Posé Mirage 2000 Trains Non Sortis Cambrai

URL: https://www.youtube.com/watch?v=vBRadALkXK0

Content:

- Different warning tones plus HUD at landing

12.6.16 Y16: Hellenic Air Force Mirage 2000EG

URL: https://www.youtube.com/watch?v=qD3Atv7uF5g

Content:

- Around 14 min: Aero braking and HUD display approach/ground

12.6.17 Y17: Fighter Jets: Engineered to Push Pilots to the Limit | Documentary

URL: https://www.youtube.com/watch?v=QeEhiRkg6BA

Content:

- In general: the evolution of the Ouragan, M2000 and Rafale
- Around ca. 19:22 and 20:34: footage of -5 cockpit instruments from another simulator
- Around ca. 24:00: footage of TGP display

12.6.18 Y18: Mon simulateur de Mirage 2000

URL: https://www.youtube.com/watch?v=qak1jI_aRAM

Content:

- Video of a home-made Mirage 2000-5F cockpit with a majority of original parts.
- Close up video sequences of all panels with angles making it possible to also see depth

PS: In the show notes there is a link to the author's Tiktok channel, where there are more relevant shorts.

13 Abbreviations and Jargon

13.1 French Abbreviations (Glossary)

Abbreviation	Meaning	English
A / AR	arrête	off
AST	assaut en suivi de terrain	terrain-following assault mode
AU	automatique	automatic
BFR	Basse FRéquence	Low Frequency (radar pulse frequency)
CAS	cannon air-sol	cannon in air-ground mode
CCPI	Calcul Continu du Point d'Impact	CCIP (Continuously Computed Impact Point)
CCPL	Calcul Continu du Point de Largage	CCRP (Continuously Computed Release Point)
ENT	Entrelacé	Interleaved (radar pulse frequency)
D	droite	right
DIRAV	Dirigeabilité Roue Avant	nose wheel steering(NWS)
G	gauche	left
HFR	Haute FRéquence	High Frequency (radar pulse frequency)
ISG	indicateur sphérique gyroscopique	attitude indicator
JVN	jumelles de vision nocturnes	night vision goggles
LL	lance leurres	decoy dispenser
M	marche	on
NOSA	navigateur officier systèmes d'armes	weapon systems officer
OPEX	opération extérieure	foreign operations
P	prêt	ready
PA	pilote automatique	autopilot
PC	post-combustion	reheat / afterburner
PCA	poste de commande armement	
PCN	poste de commande navigation	
PCR	poste de commande radar	
PGS	plein gaz sec	full military power (without reheat)
PIC	Poursuite sur Information Continue	STT = Single Target Track

PID	Poursuite sur Information Discontinue	TWS = Track While Scanning
PLA	point de largage de l'armement	weapons release point
PPA	poste préparation armement	
PSM	poste sélecteur de modes	
RTBA	réseau très basse altitude	military low-flying areas
S	selectionné	selected
SA	semi-automatique	semi-automatic
SCALP	système de croisière conventionnel autonome à longue portée	conventional cruise missile
SDT	suivi de terrain	terrain following
TBA	très basse altitude	very low altitude
VTB	visualisation tête basse	HDD - head-down display
VTM	visualisation tête moyenne	n/a - head level display
VTH	visualisation tête haute	HUD - head-up display

13.2 French Military Aviation Jargon

Jargon	Meaning
passage de dos	inverted flying

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